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LEE et al.(10) **Pub. No.: US 2025/0285915 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SEMICONDUCTOR INTERCONNECTION
STRUCTURES AND METHODS OF
FORMING THE SAME****Publication Classification**

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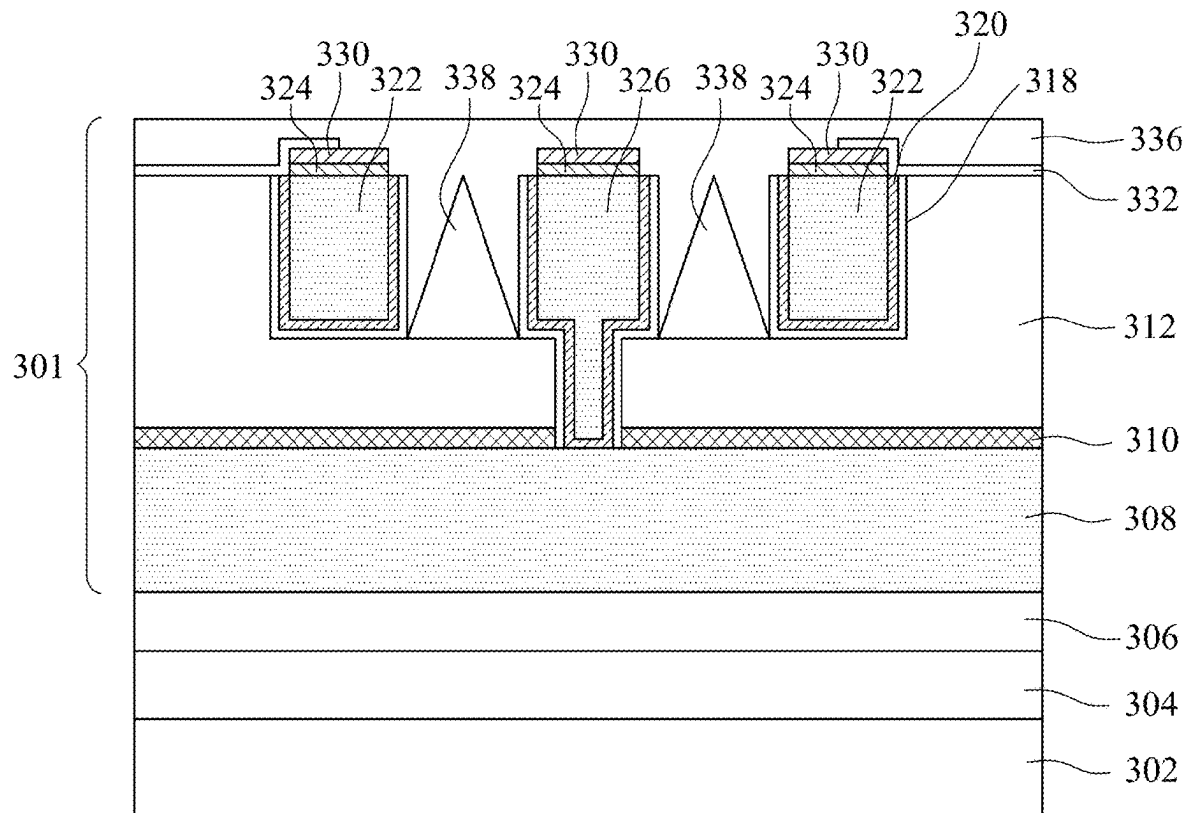
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18, 2022, now Pat. No. 12,272,597.(60) Provisional application No. 63/212,002, filed on Jun.
17, 2021.

(57)

ABSTRACT

An interconnection structure includes a first dielectric layer, a first conductive feature, a first liner layer, a second conductive feature, a second liner layer, and an air gap. The first conductive feature is disposed in the first dielectric layer. The first liner layer is disposed between the first conductive feature and the first dielectric layer. The second conductive feature penetrates the first dielectric layer. The second liner layer is disposed between the second conductive feature and the first dielectric layer. The air gap is disposed in the first dielectric layer between the first liner layer and the second liner layer. The first liner layer and the second liner layer include metal oxide, metal nitride, or silicon oxide doped carbide.

300

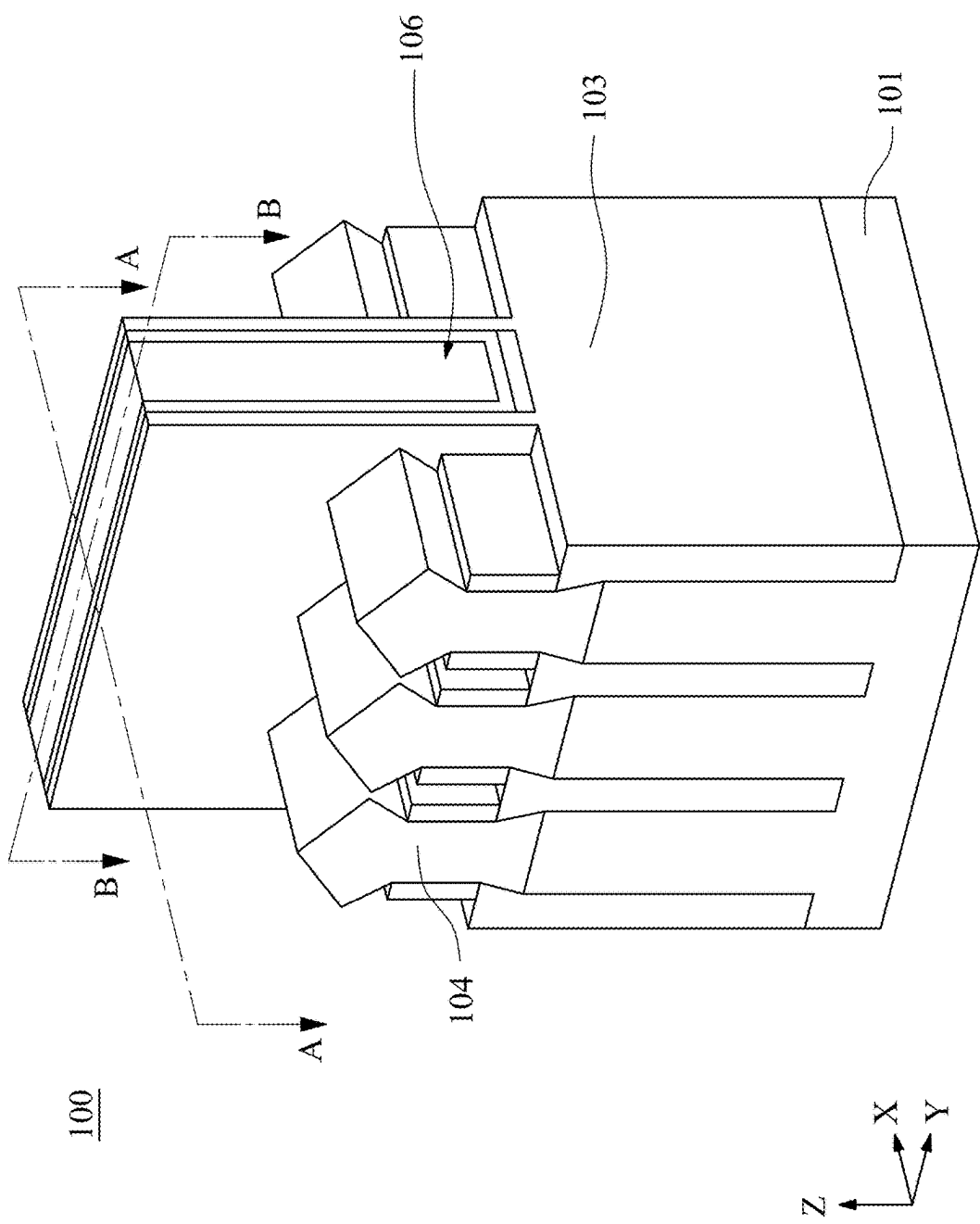


Fig. 1

100

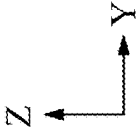
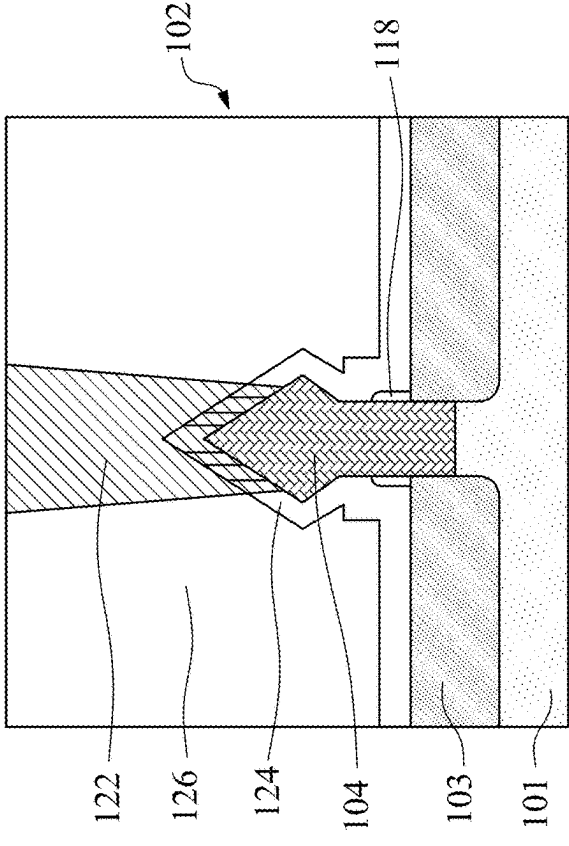


Fig. 2B

100

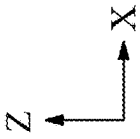
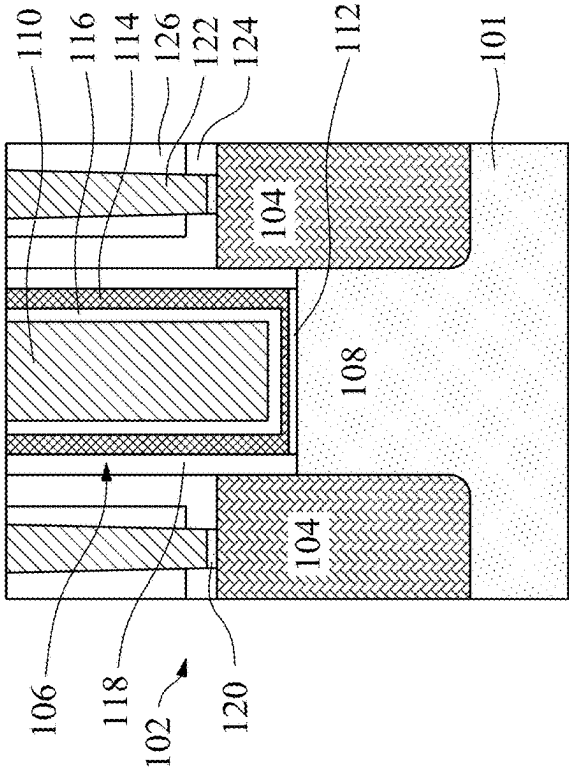


Fig. 2A

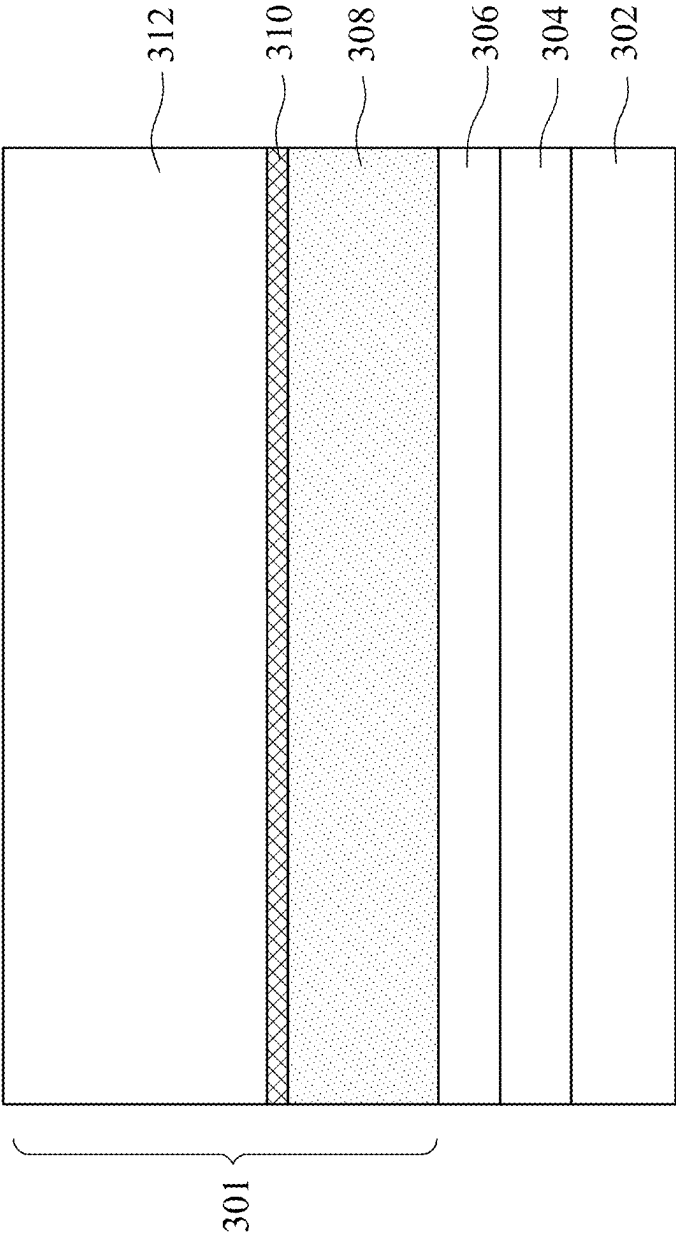


Fig. 3A

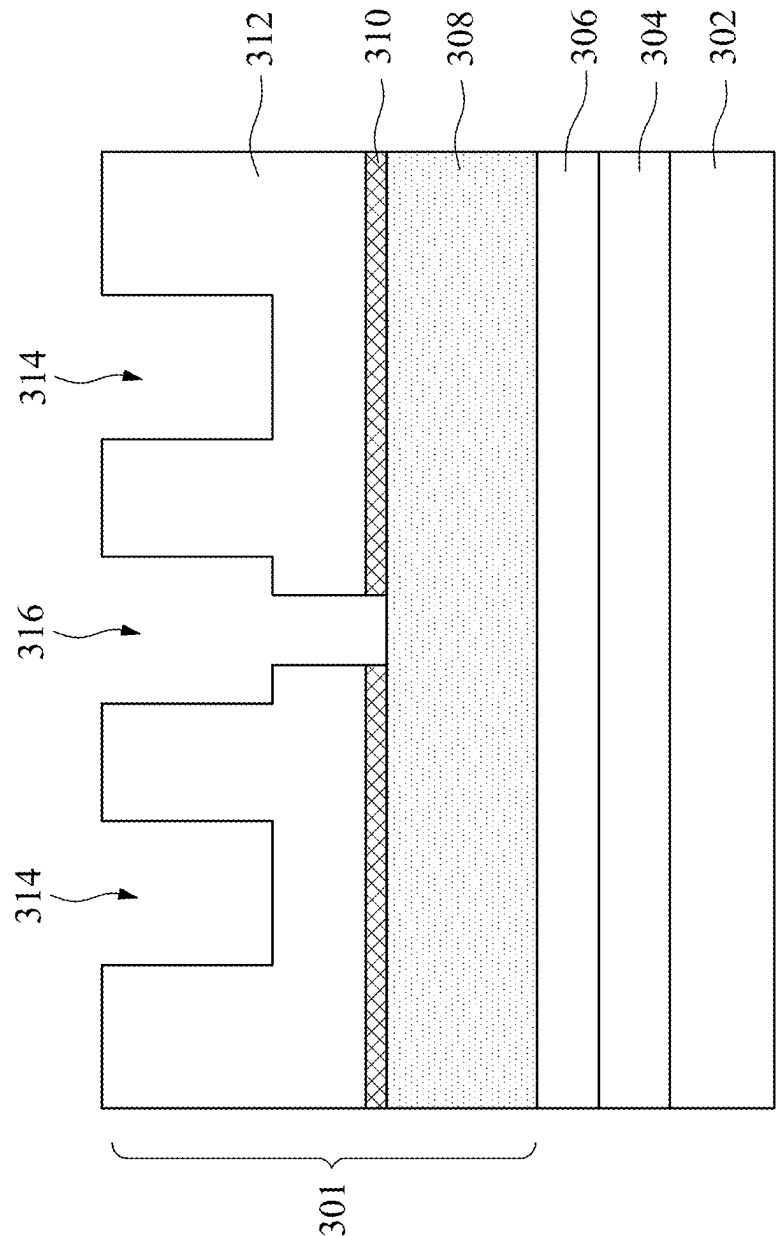


Fig. 3B

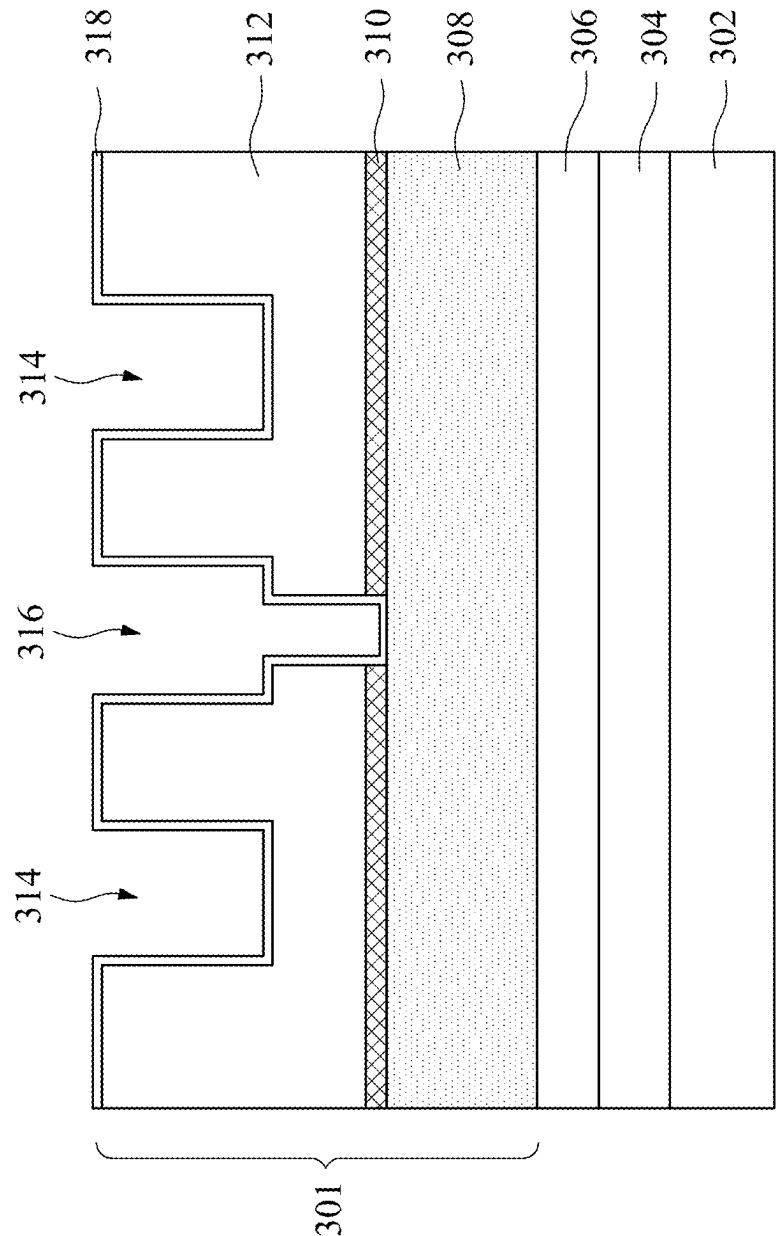


Fig. 3C

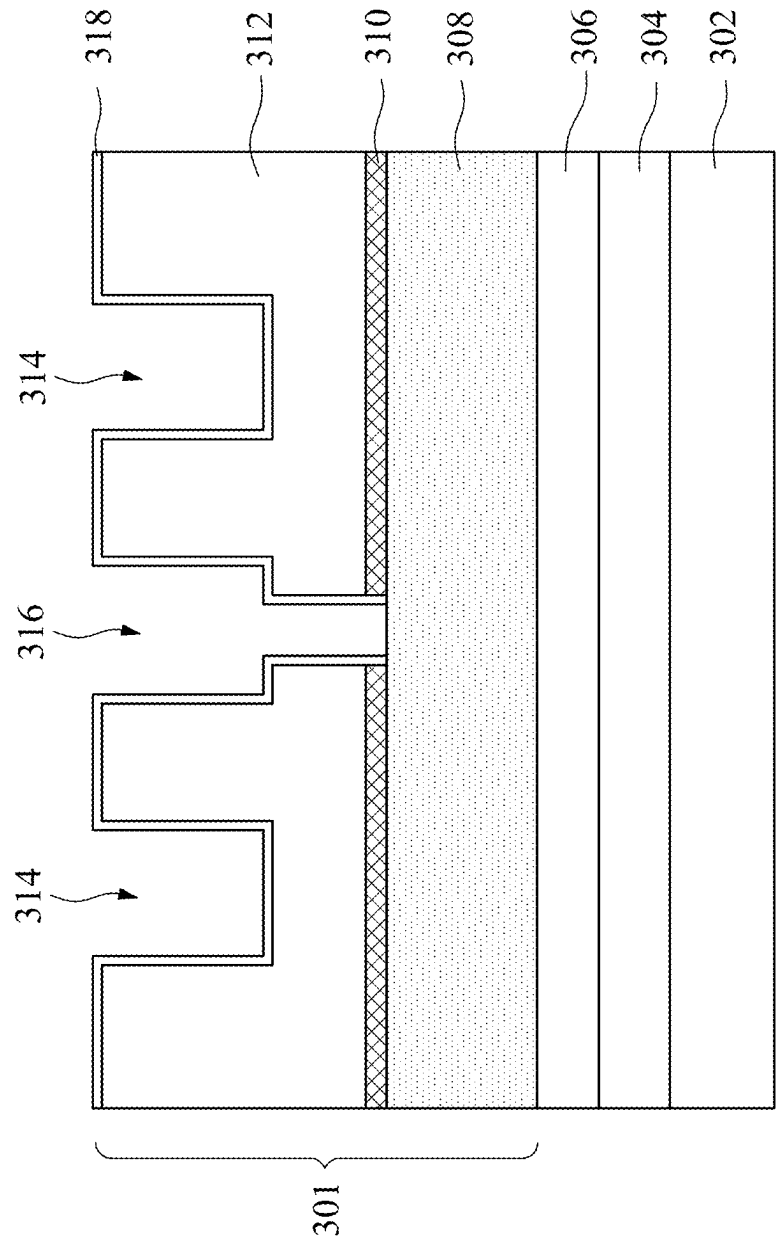


Fig. 3D

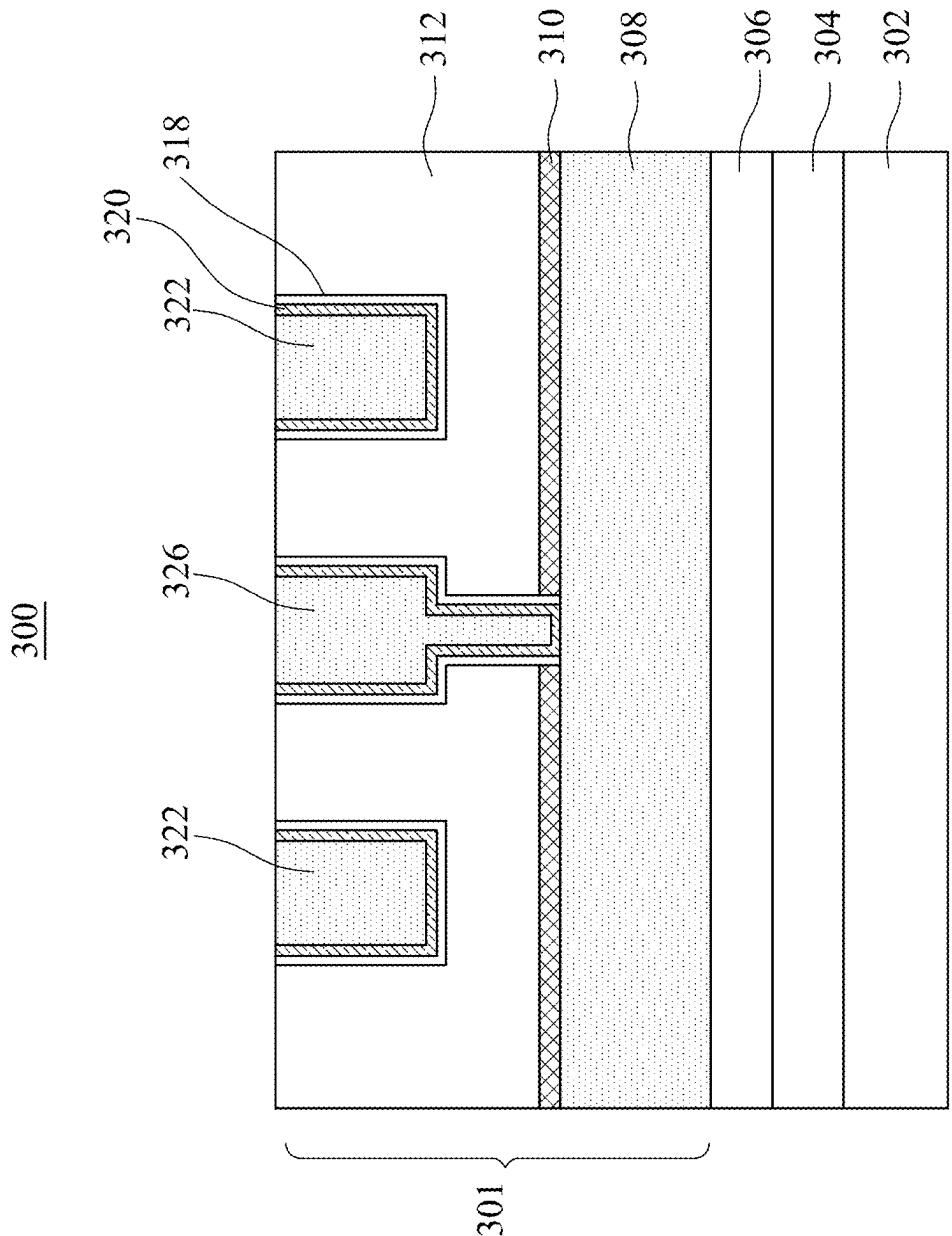


Fig. 3E

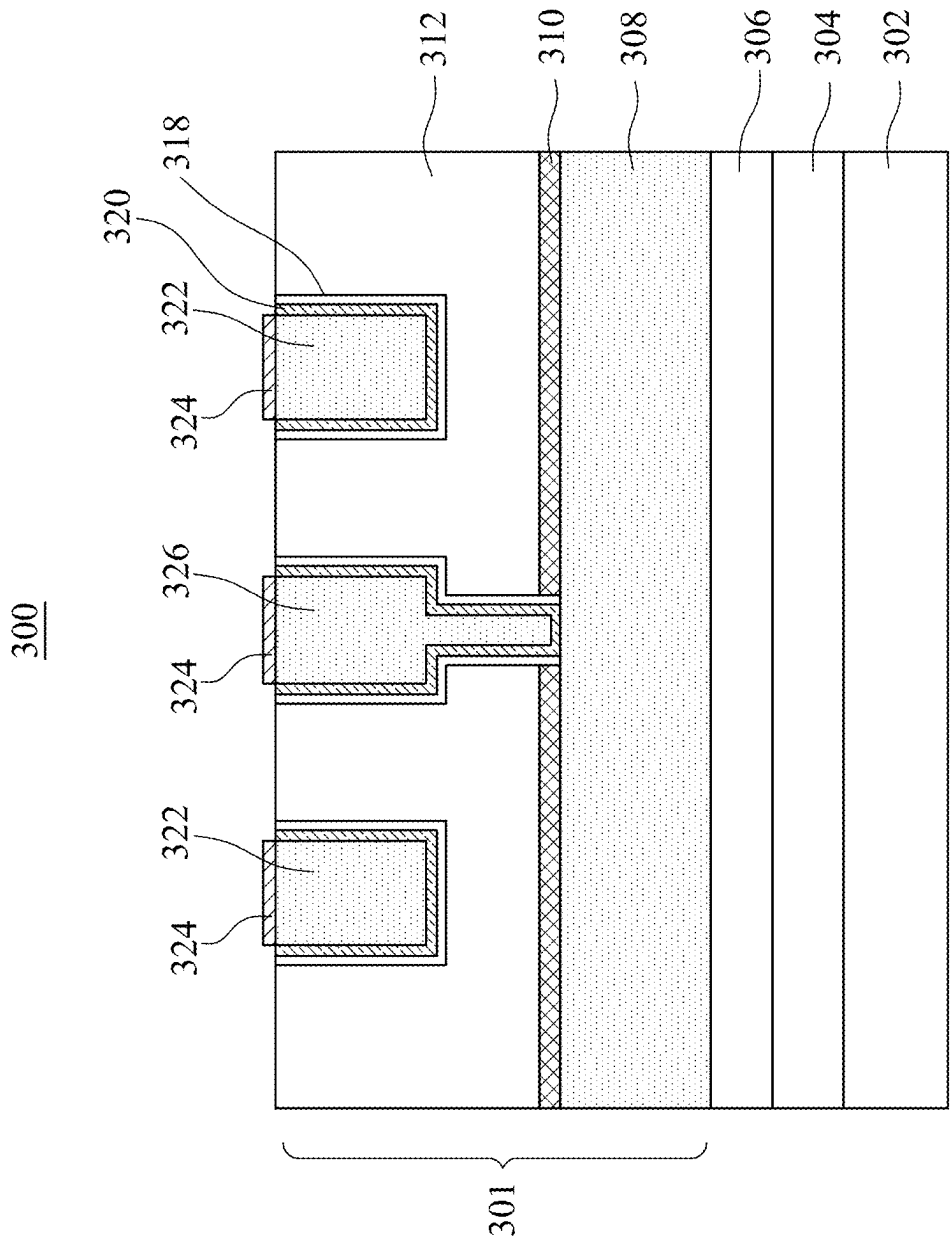


Fig. 3F

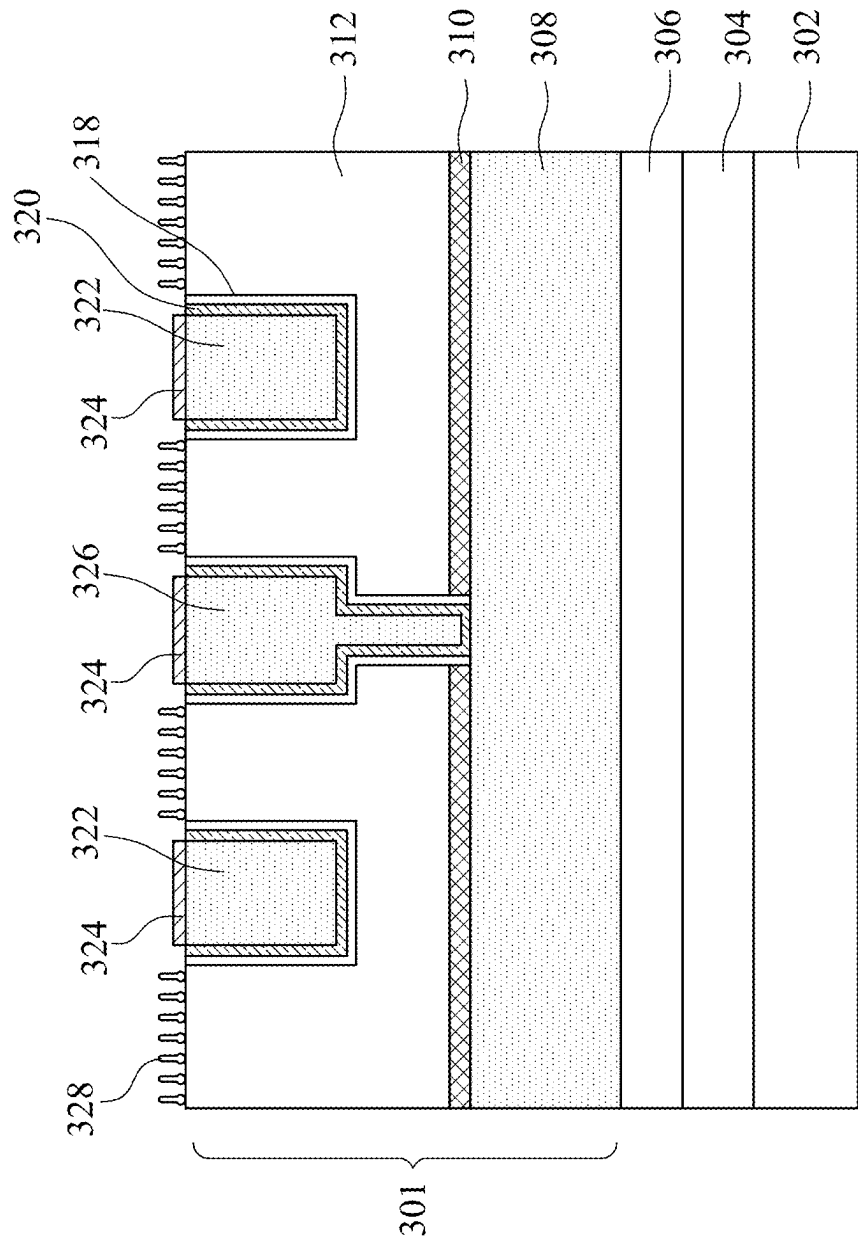


Fig. 3G

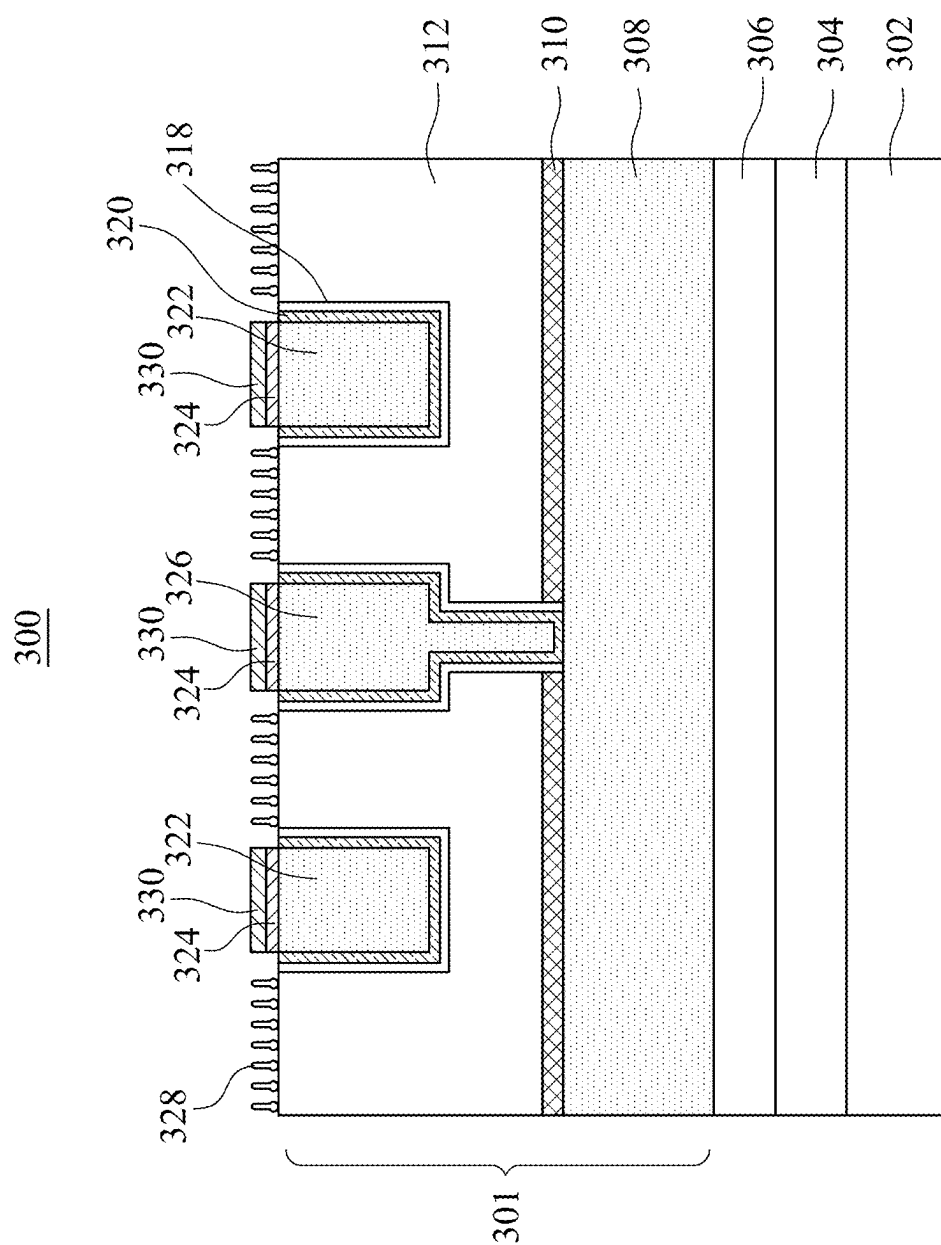


Fig. 3H

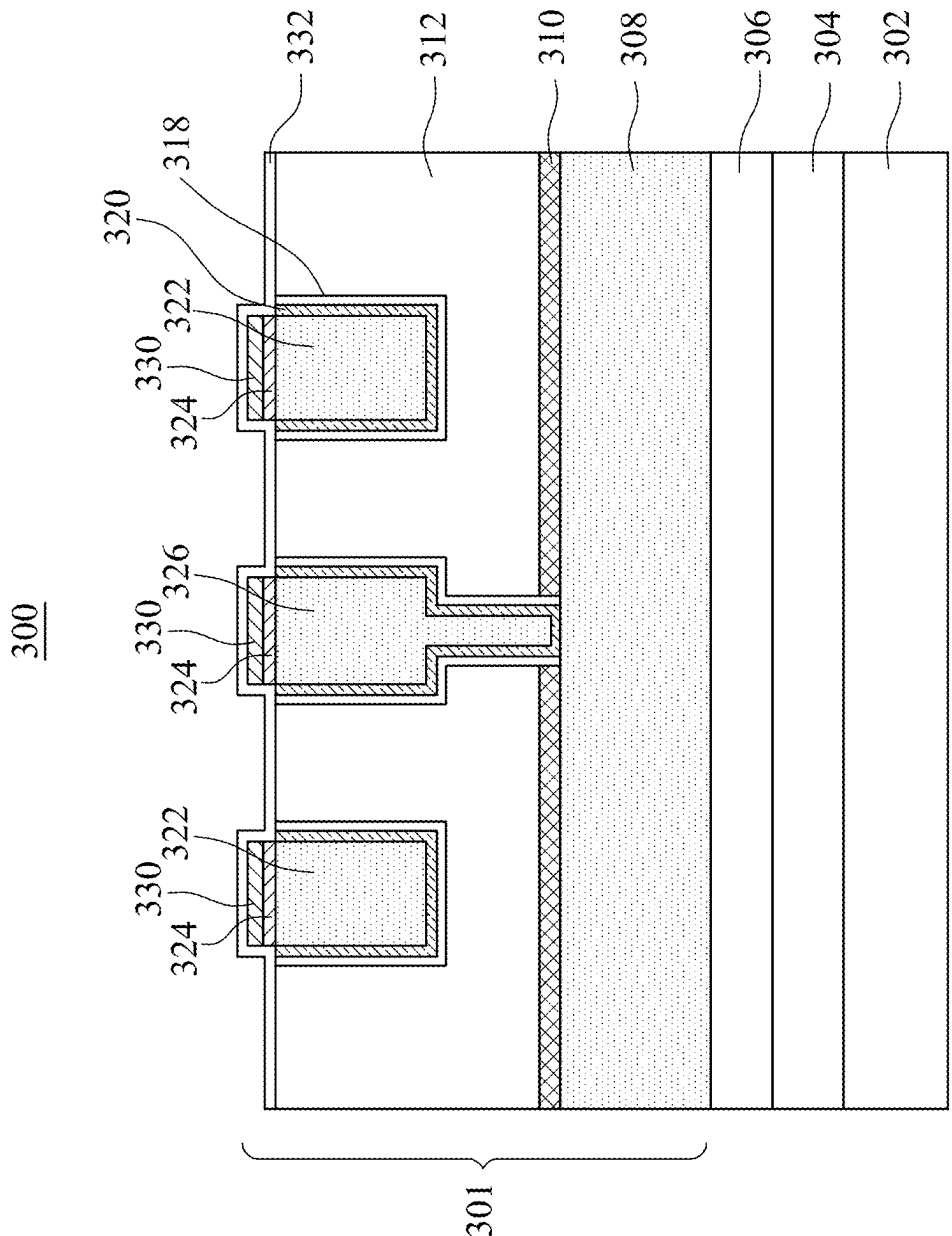


Fig. 3I

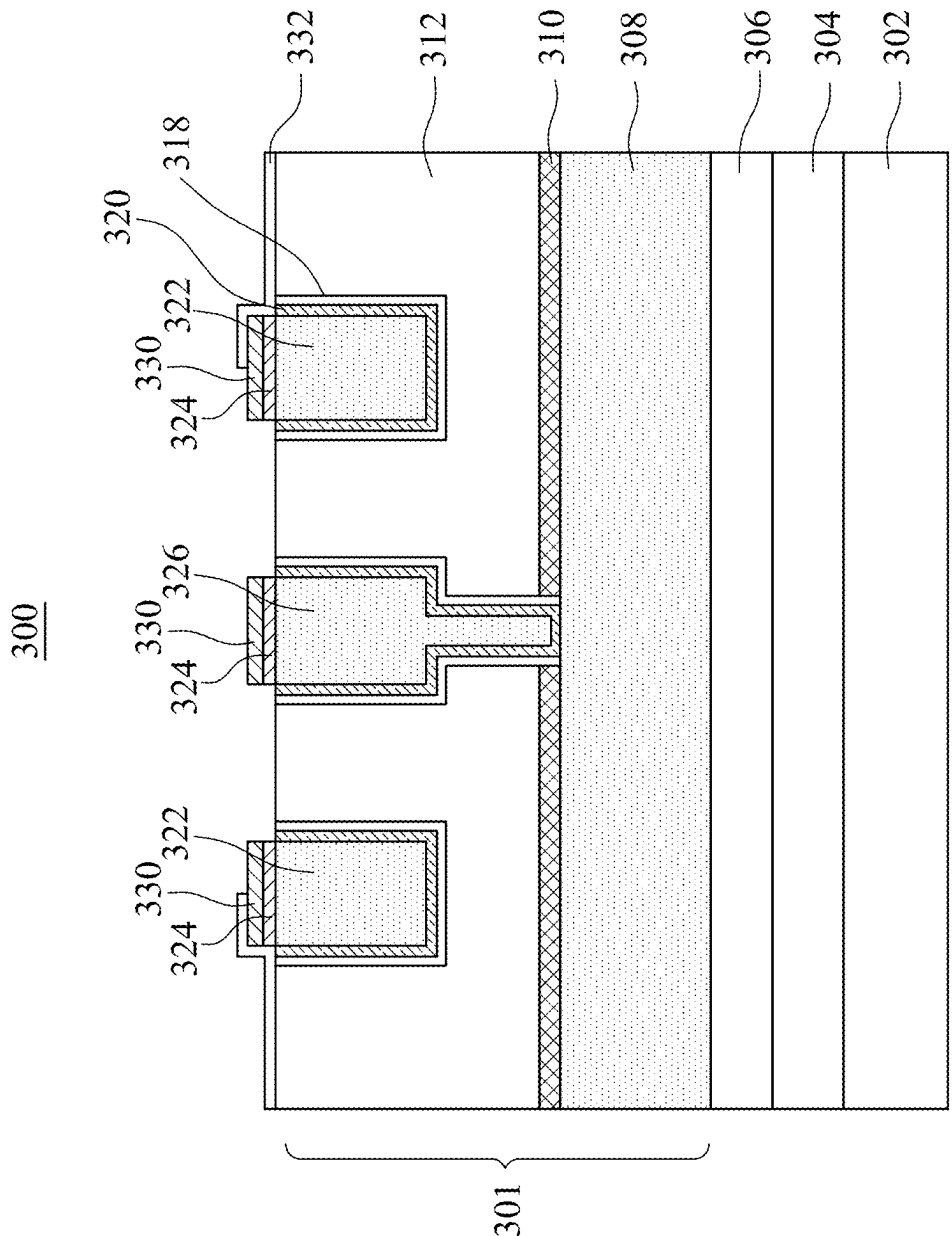


Fig. 3J

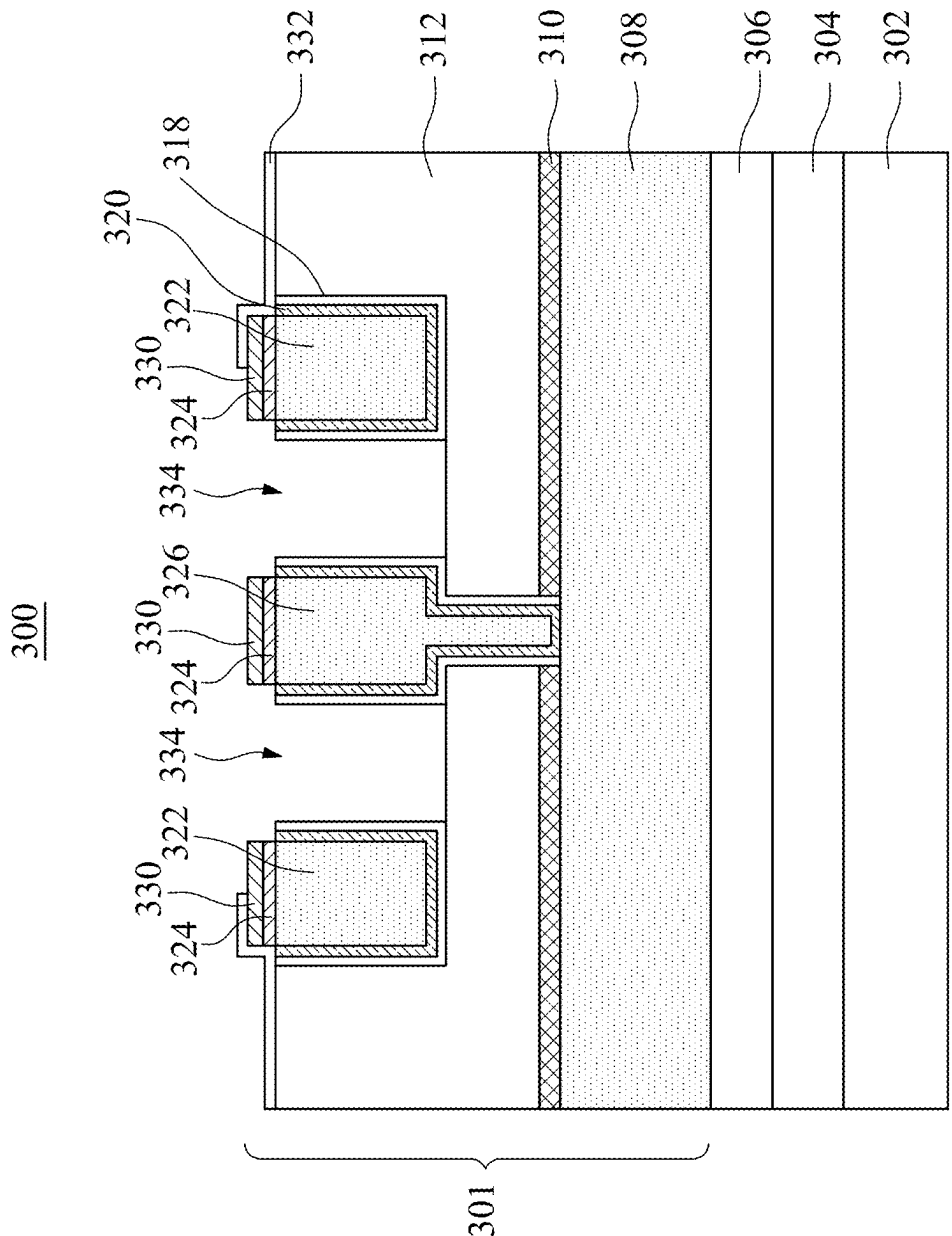


Fig. 3K

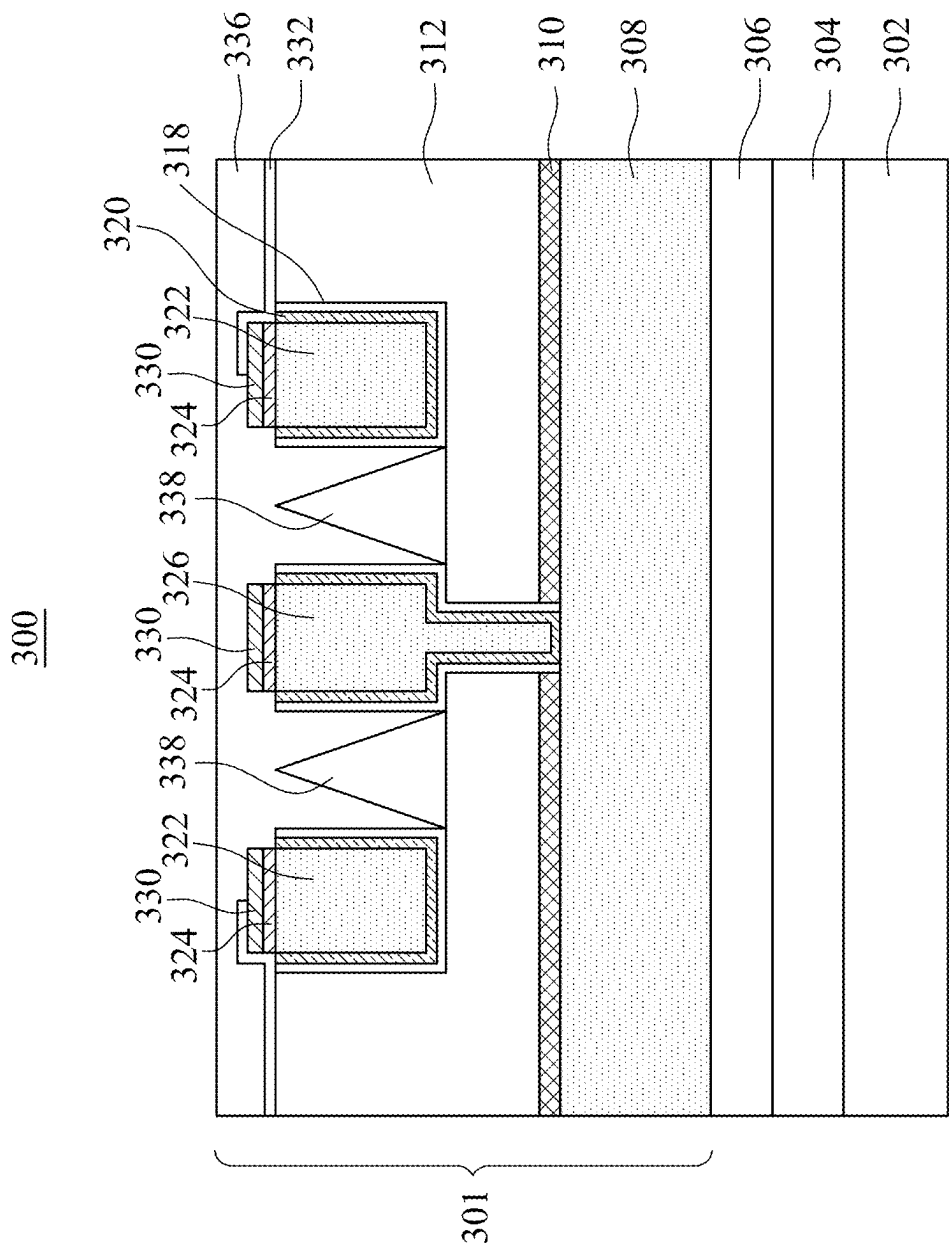


Fig. 3L

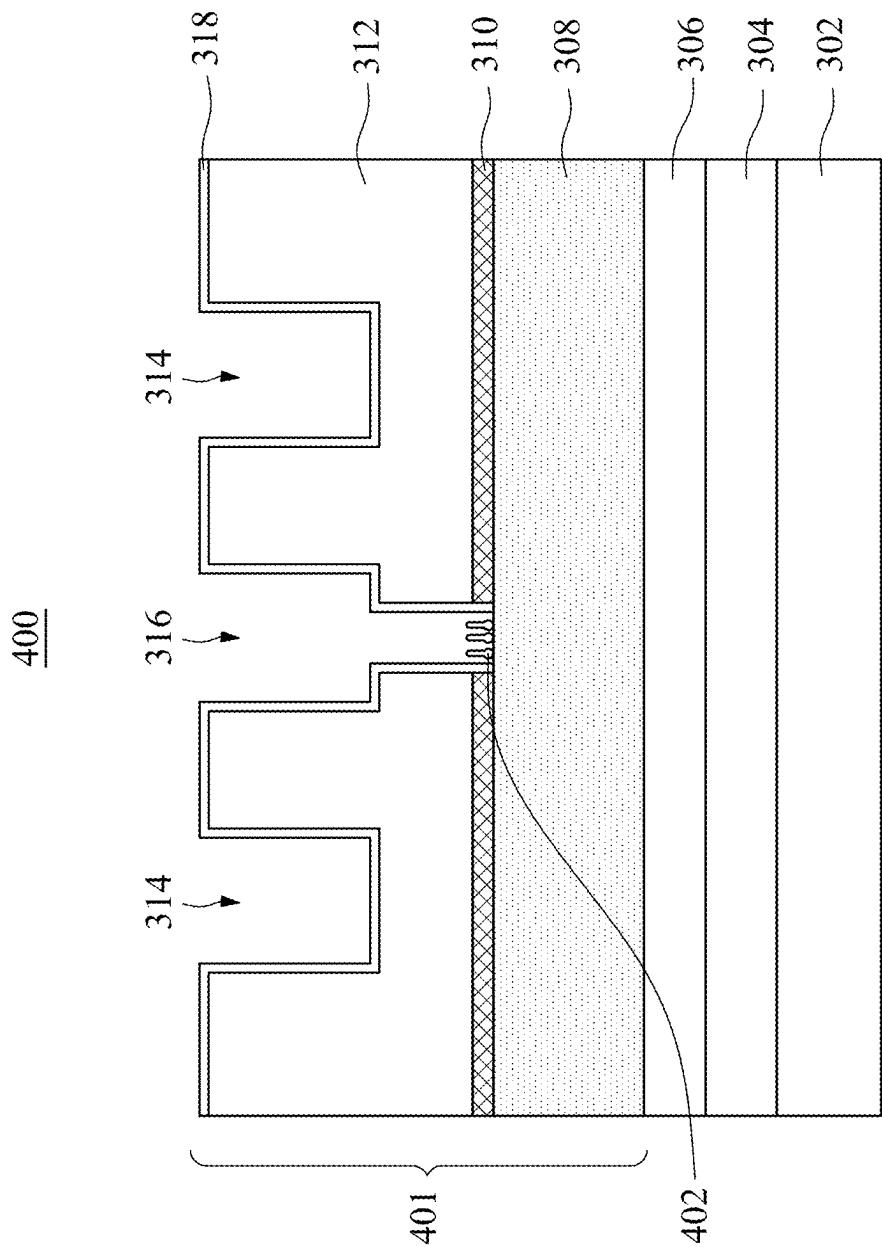


Fig. 4A

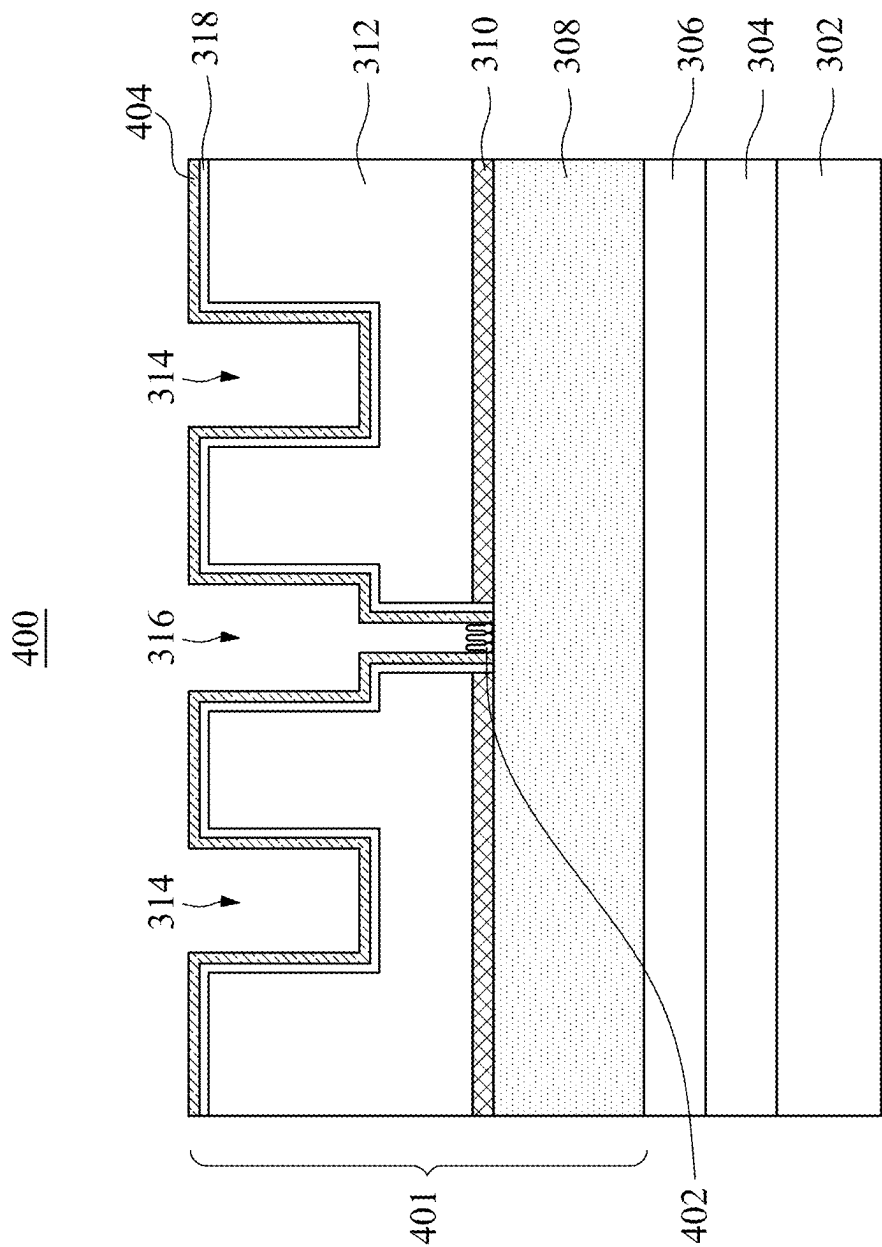


Fig. 4B

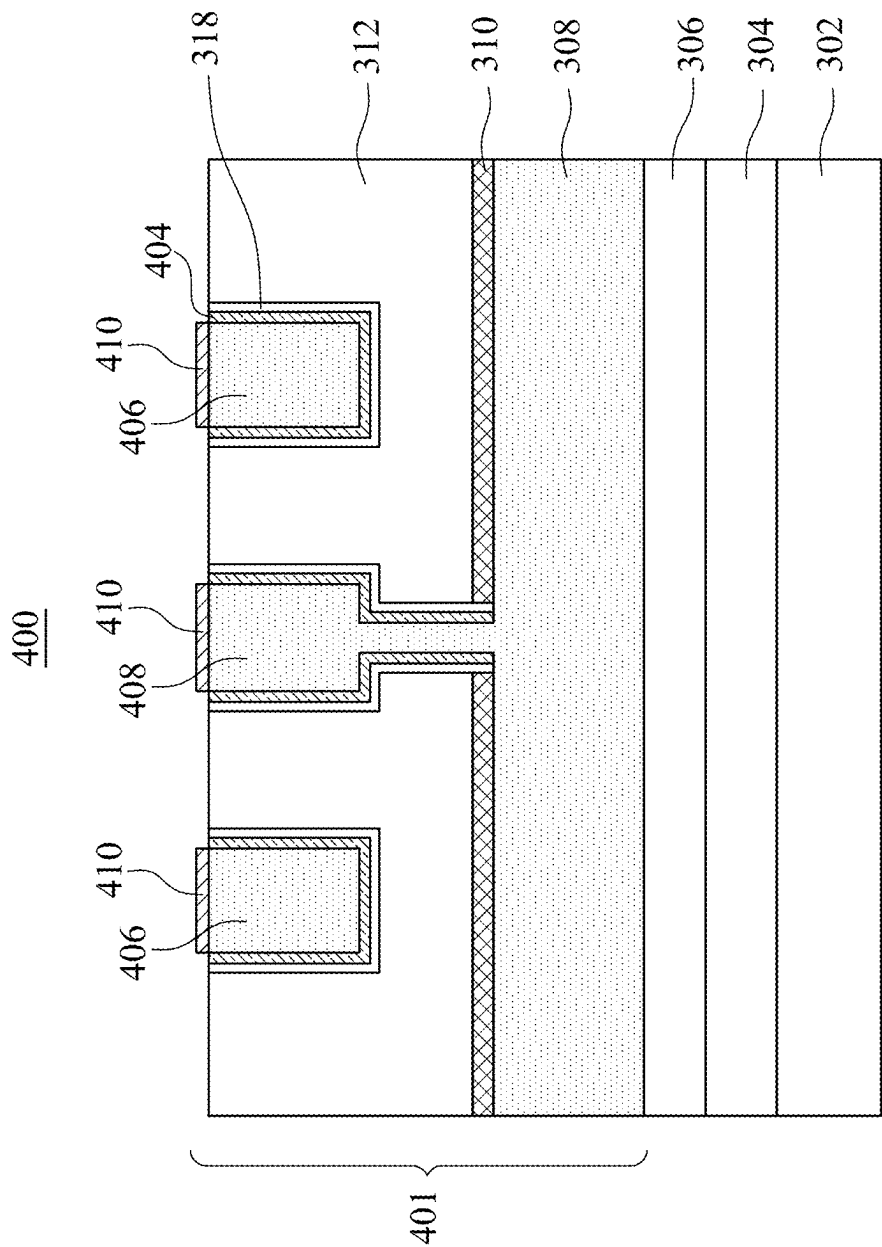


Fig. 4C

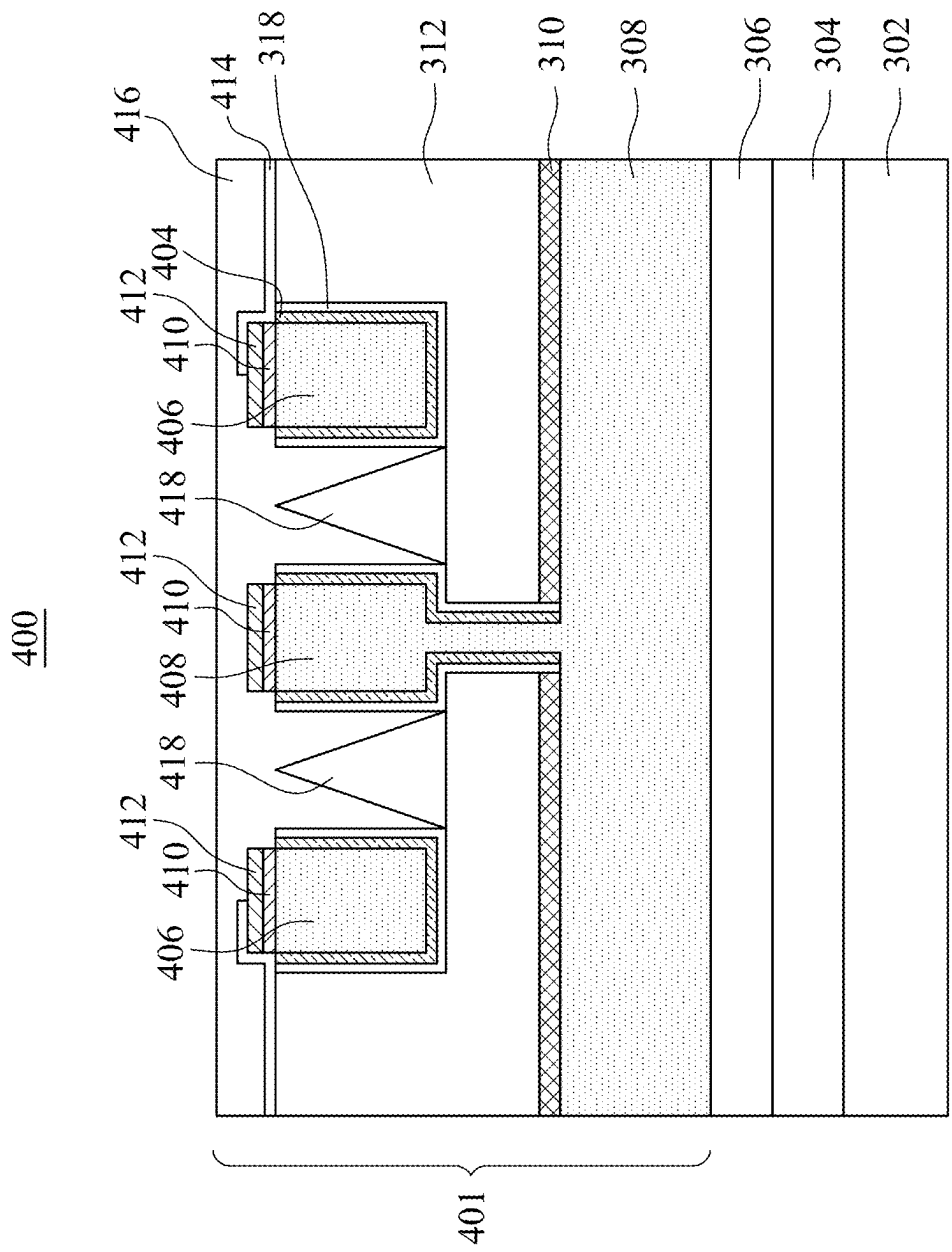


Fig. 4D

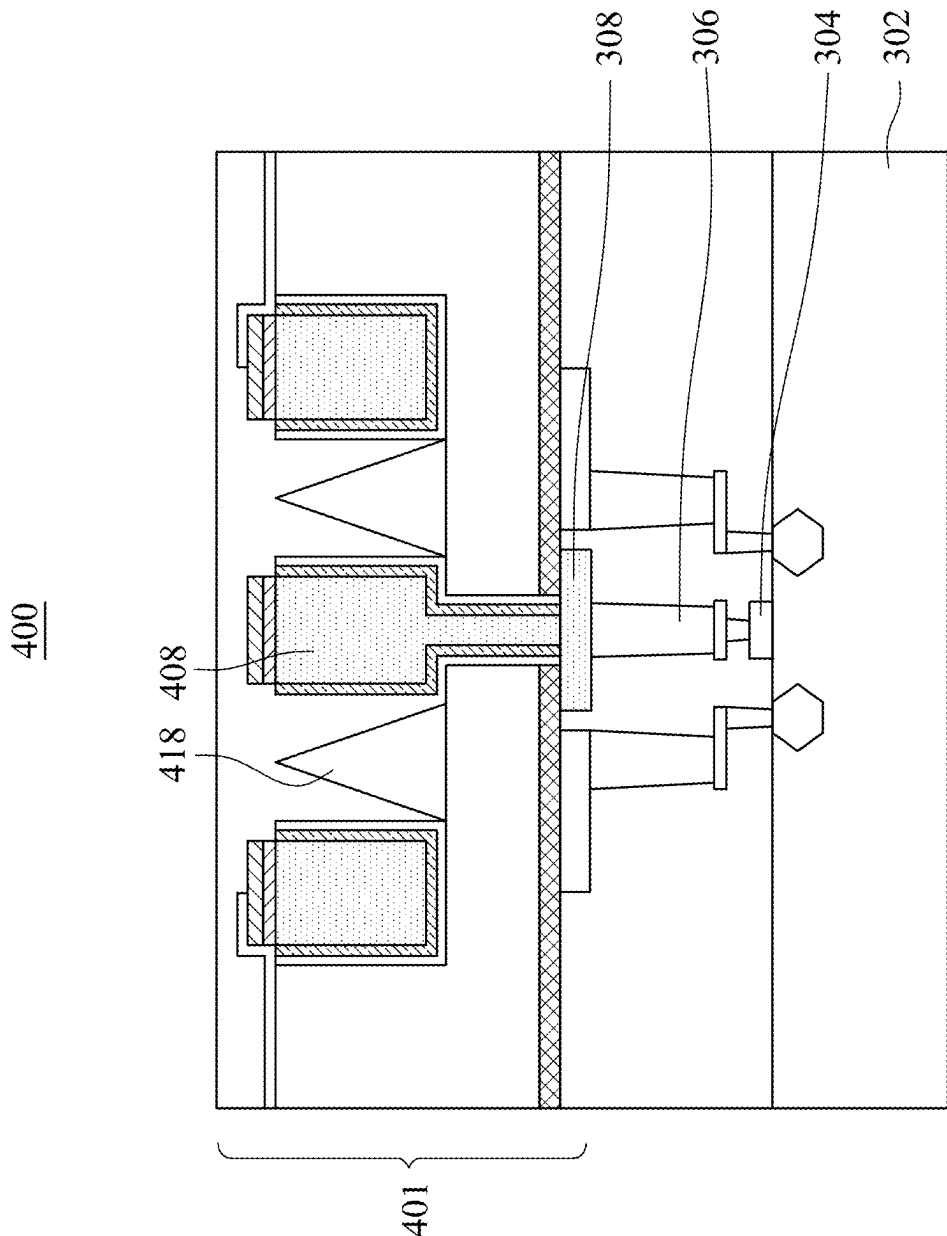


Fig. 4E

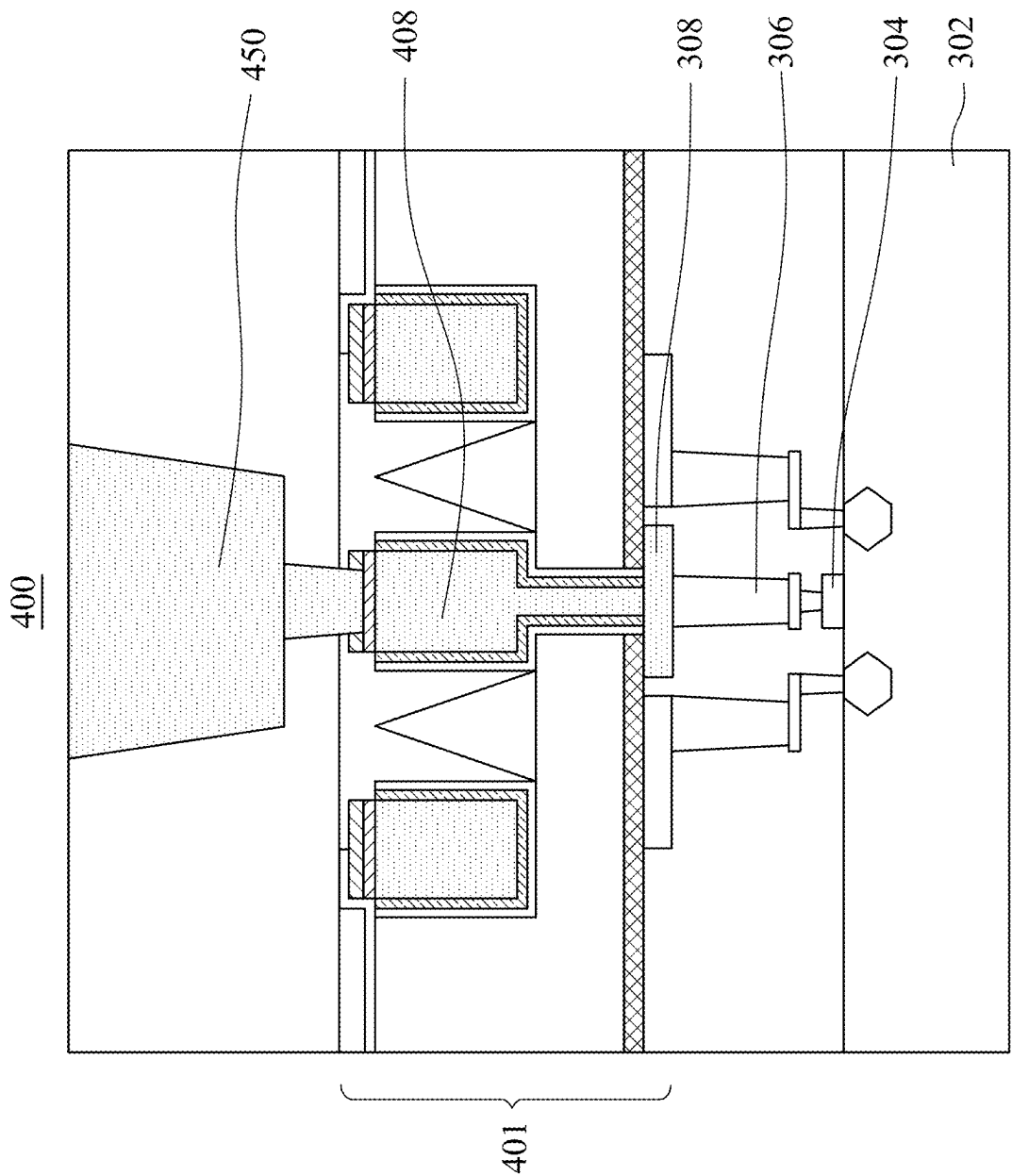


Fig. 4F

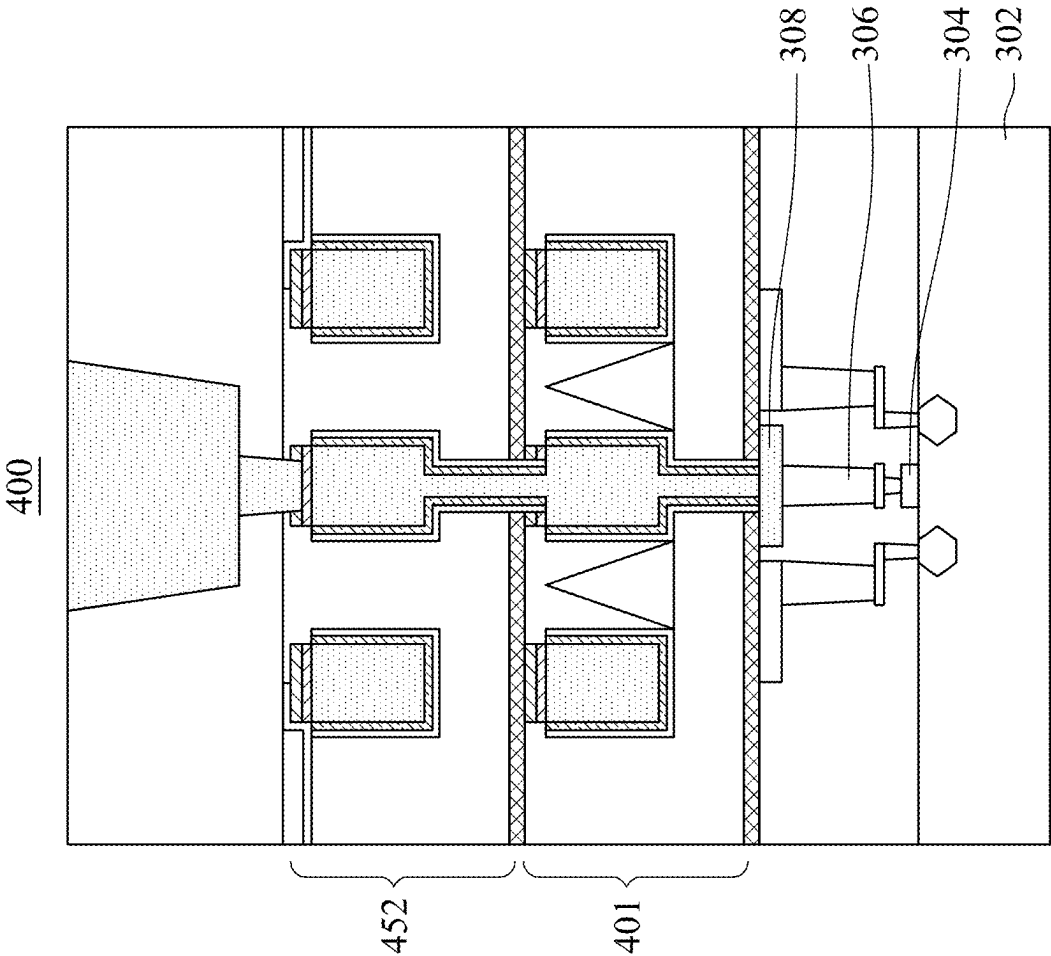


Fig. 4G

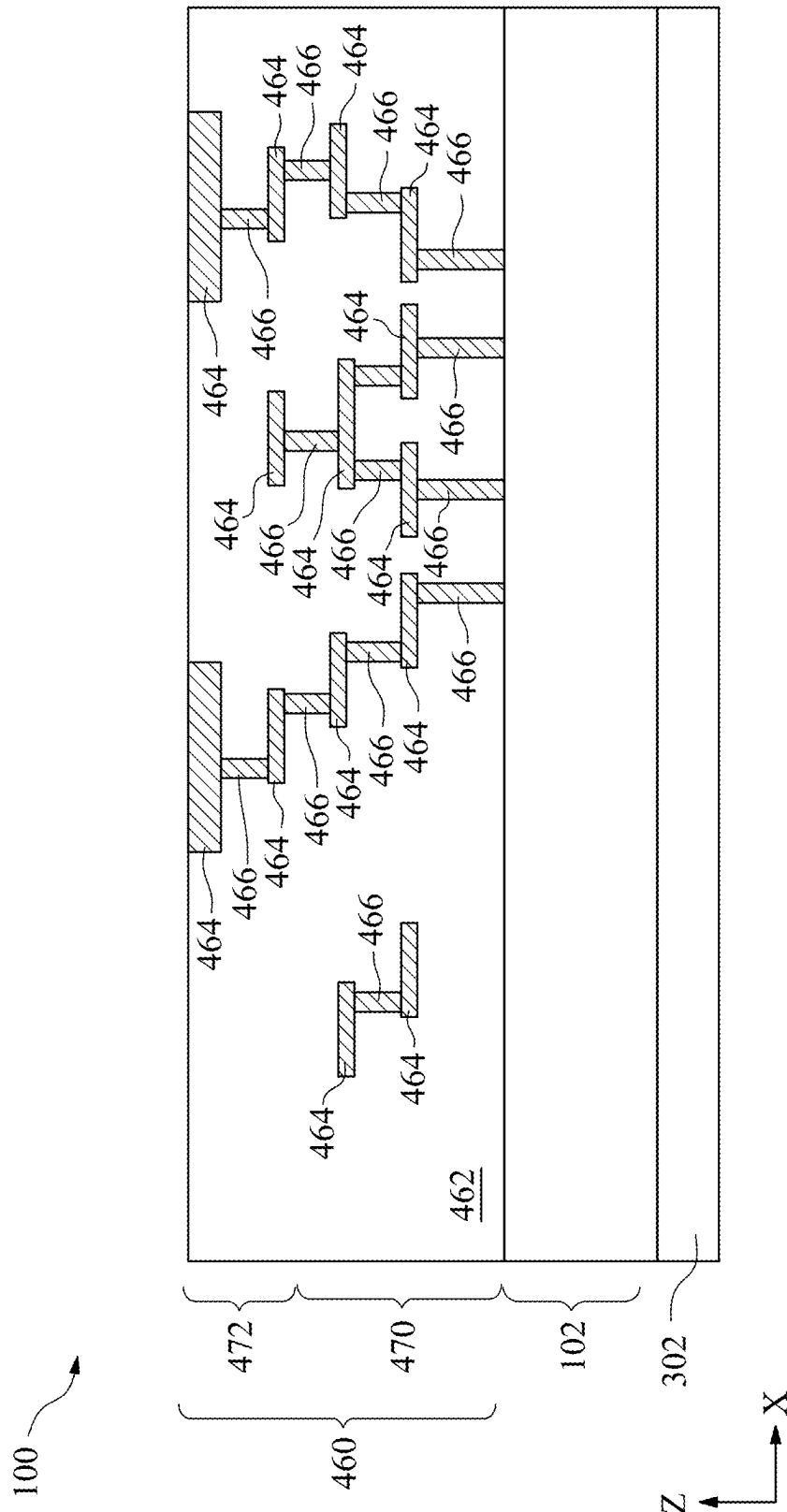


Fig. 4H

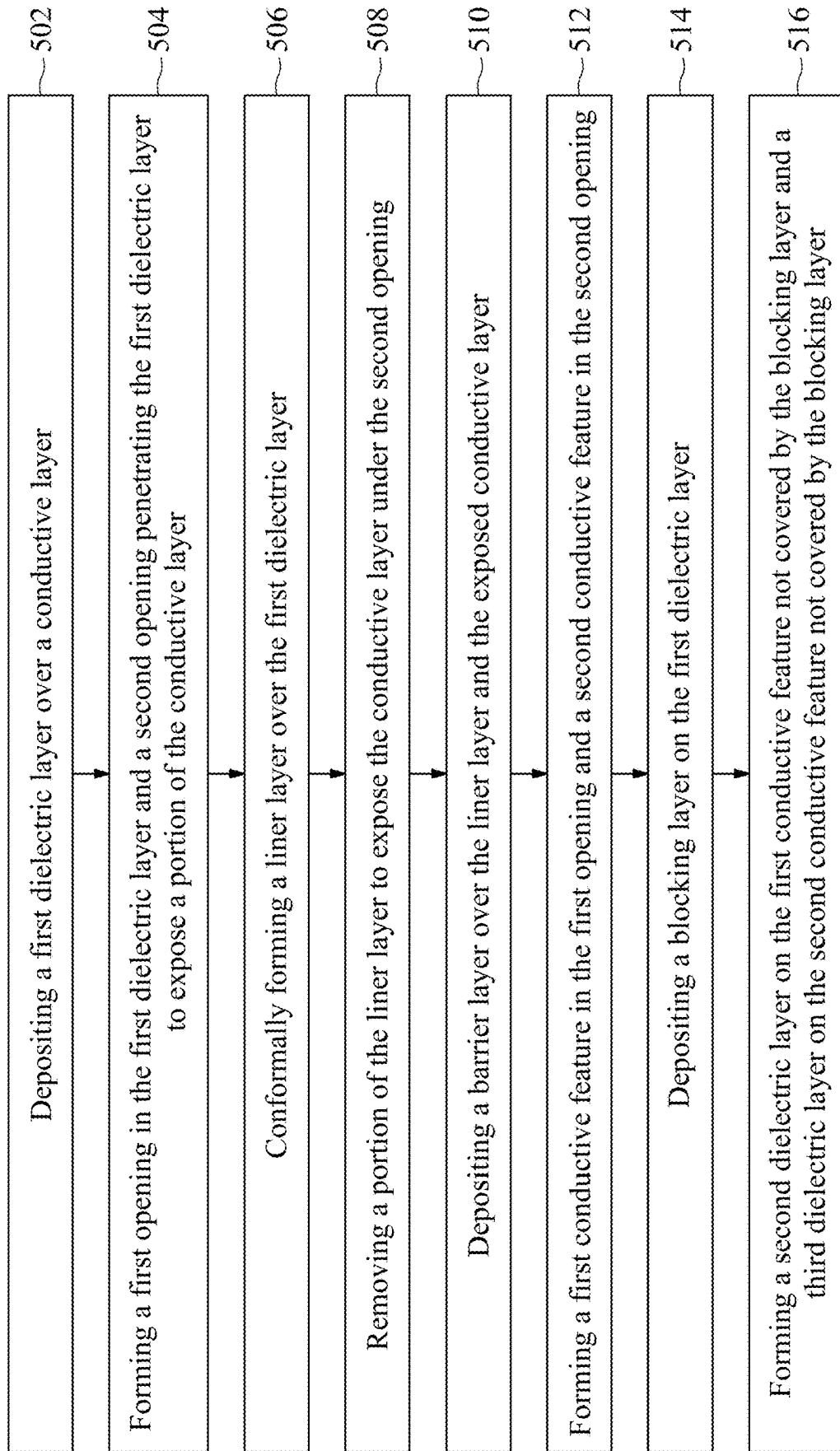
500

Fig. 5

600

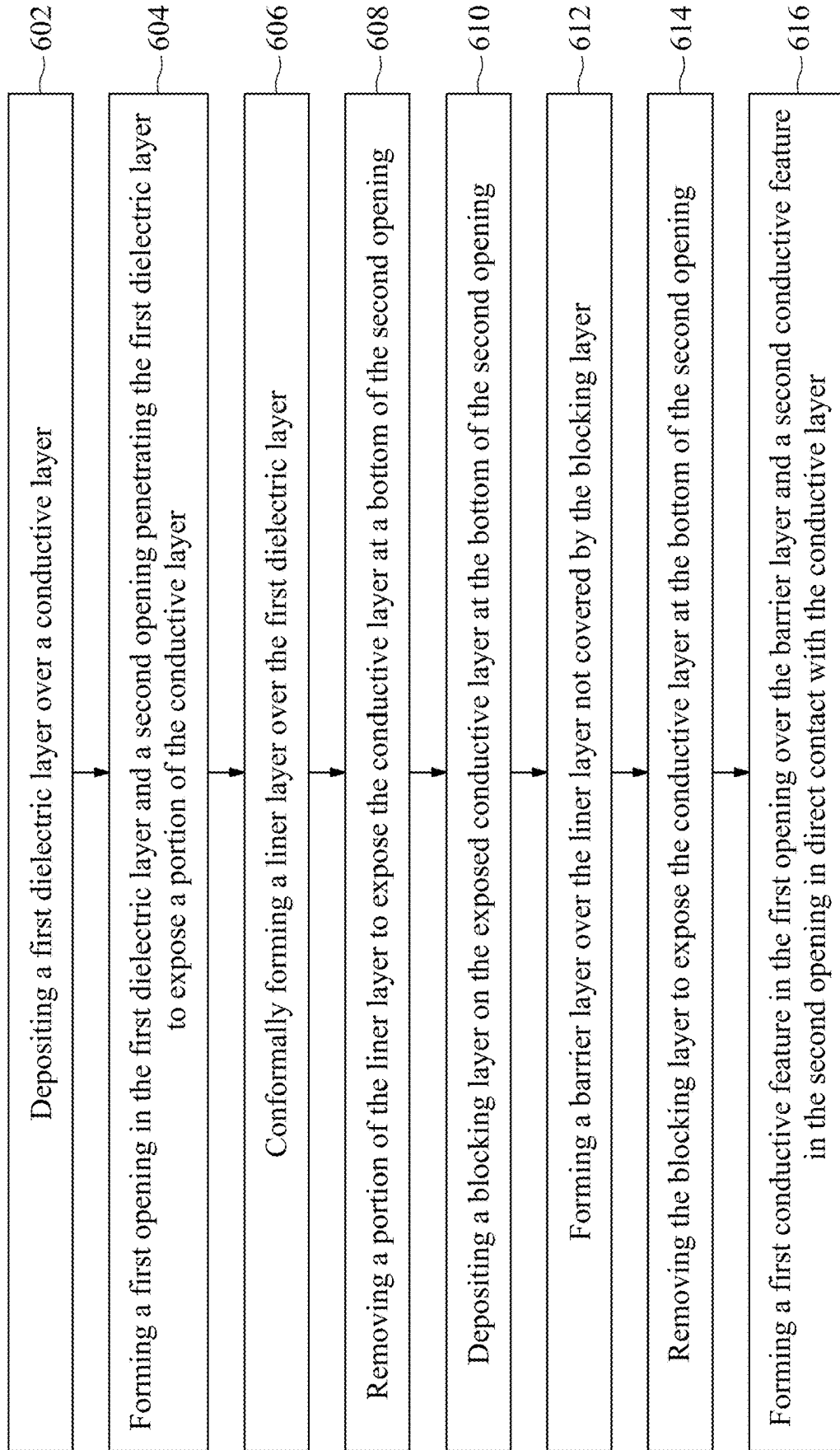


Fig. 6

SEMICONDUCTOR INTERCONNECTION STRUCTURES AND METHODS OF FORMING THE SAME

CROSS REFERENCE

[0001] This application is a Divisional application of U.S. patent application Ser. No. 17/698,743 entitled “Semiconductor Interconnection Structures and Methods of Forming the Same,” filed Mar. 18, 2022, which claims priority to U.S. Provisional Application Ser. No. 63/212,002 filed Jun. 17, 2021. Each of the aforementioned applications is incorporated by reference in its entirety.

BACKGROUND

[0002] As the semiconductor industry introduces new generations of integrated circuits (IC) having higher performance and more functionality, the density of the elements forming the ICs increases, while the dimensions, sizes and spacing between components or elements are reduced. The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area.

[0003] For manufacturing different conductive layers on the substrate, various structures and manufacturing methods are utilized to form the interconnection structures between the conductive layers. However, the integrated fabrication also brings out some issues, such as reliability, high capacitance, or high resistance. Therefore, there is a need in the art to provide improved devices or methods that can address the issues mentioned above.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 is a perspective view of one of the various stages of manufacturing a semiconductor device structure, in accordance with some embodiments.

[0006] FIGS. 2A-2B are cross-sectional side views of one of various stages of manufacturing a semiconductor device structure, in accordance with some embodiments.

[0007] FIGS. 3A-3L are cross-sectional side views of various stages of manufacturing a semiconductor structure, in accordance with some embodiments.

[0008] FIGS. 4A-4H are cross-sectional side views of various stages of manufacturing another semiconductor structure, in accordance with some embodiments.

[0009] FIG. 5 is a flow chart of a method for manufacturing a semiconductor interconnection structure in accordance with some embodiments.

[0010] FIG. 6 is a flow chart of another method for manufacturing a semiconductor interconnection structure in accordance with some embodiments.

DETAILED DESCRIPTION

[0011] The following disclosure provides many different embodiments, or examples, for implementing different fea-

tures of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “on,” “top,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] FIG. 1 is a perspective view of one of the various stages of manufacturing a semiconductor device structure **100**, in accordance with some embodiments. As shown in FIG. 1, the semiconductor device structure **100** includes a substrate **101** having at least a plurality of devices formed thereover. The devices, such as transistors, diodes, imaging sensors, resistors, capacitors, inductors, memory cells, a combination thereof, and/or other suitable devices, may be formed on the substrate **101**. In some embodiments, the interconnection structures may be formed on or below the devices.

[0014] FIGS. 2A-2B are cross-sectional side views of various stages of manufacturing the semiconductor device structure **100**, in accordance with some embodiments. FIG. 2A is a cross-sectional side view of the semiconductor device structure **100** taken along line A-A of FIG. 1, and FIG. 2B is a cross-sectional side view of the semiconductor device structure **100** taken along line B-B of FIG. 1. The line A-A of FIG. 1 extends along a direction that is substantially perpendicular to the longitudinal direction of a gate stack **106**, and the line B-B of FIG. 1 extends along the longitudinal direction of the gate stack **106**. As shown in FIGS. 2A and 2B, the semiconductor device structure **100** includes the substrate **101**, and one or more devices **102** are formed on the substrate **101**. The interconnection structures may be formed over the devices **102**.

[0015] The substrate **101** may be a semiconductor substrate. In some embodiments, the substrate **101** includes a crystalline semiconductor layer on at least the surface of the substrate **101**. The substrate **101** may include a crystalline semiconductor material such as, but not limited to silicon (Si), germanium (Ge), silicon germanium (SiGe), gallium arsenide (GaAs), indium antimonide (InSb), gallium phosphide (GaP), gallium antimonide (GaSb), indium aluminum arsenide (InAlAs), indium gallium arsenide (InGaAs), gallium antimony phosphide (GaSbP), gallium arsenic antimonide (GaAsSb), and indium phosphide (InP). In some embodiment, the substrate **101** is made of Si. In some

embodiments, the substrate **101** is a silicon-on-insulator (SOI) substrate, which includes an insulating layer (not shown) disposed between two silicon layers. In one aspect, the insulating layer is an oxygen-containing material, such as an oxide.

[0016] The substrate **101** may include various regions that have been suitably doped with impurities (e.g., p-type or n-type impurities). The dopants are, for example phosphorus for an n-type fin field effect transistor (FinFET) and boron for a p-type FinFET.

[0017] As described above, the devices **102** may be any suitable devices, such as transistors, diodes, imaging sensors, resistors, capacitors, inductors, memory cells, or a combination thereof. In some embodiments, the devices **102** are transistors, such as planar field effect transistors (FETs), FinFETs, nanostructure transistors, or other suitable transistors. The nanostructure transistors may include nanosheet transistors, nanowire transistors, gate-all-around (GAA) transistors, multi-bridge channel (MBC) transistors, or any transistors having the gate electrode surrounding the channels. An example of the device **102** formed between the substrate **101** and the interconnection structures (such as the interconnection structure **200** shown in FIG. 3A-3L or 4A-4D) may be a FinFET or a nanostructure, which is shown in FIGS. 2A and 2B. An exemplary device **102** may include source/drain (S/D) regions **104** and a gate stack **106** disposed between the S/D regions **104** serving as source regions and the S/D regions **104** serving as drain regions. While there is only one gate stack **106** formed on the substrate **101**, it is contemplated that two or more gate stacks **106** may also be formed on the substrate **101**. Channel regions **108** are formed between the S/D regions **104** serving as source regions and the S/D regions **104** serving as drain regions.

[0018] The S/D regions **104** may include a semiconductor material, such as Si or Ge, a III-V compound semiconductor, an II-VI compound semiconductor, or other suitable semiconductor material. Exemplary S/D region **104** may include, but are not limited to, Ge, SiGe, GaAs, AlGaAs, GaAsP, SiP, InAs, AlAs, InP, GaN, InGaAs, InAlAs, GaSb, AlP, GaP, and the like. The S/D regions **104** may include p-type dopants, such as boron; n-type dopants, such as phosphorus or arsenic; and/or other suitable dopants including combinations thereof. The S/D regions **104** may be formed by an epitaxial growth method using CVD, atomic layer deposition (ALD) or molecular beam epitaxy (MBE). The channel regions **108** may include one or more semiconductor materials, such as Si, Ge, GeSn, SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, GaN, GaP, or InP. In some embodiments, the channel regions **108** include the same semiconductor material as the substrate **101**. In some embodiments, the devices **102** are FinFETs, and the channel regions **108** are a plurality of fins each having at least three surfaces wrapped around by the gate stack **106**. In some other embodiments, the devices **102** are nanosheet transistors, and the channel regions **108** include two or more nanosheets surrounded by the gate stack **106**.

[0019] Each gate stack **106** includes a gate electrode layer **110** disposed over the channel region **108** or partially/fully surrounding the channel region **108**. The gate electrode layer **110** may be a metal-containing material such as tungsten, cobalt, aluminum, ruthenium, copper, multilayers thereof, or the like, and can be deposited by ALD, plasma enhanced chemical vapor deposition (PECVD), MBD, physical vapor

deposition (PVD), or any suitable deposition technique. Each gate stack **106** may include an interfacial dielectric layer **112**, a gate dielectric layer **114** disposed on the interfacial dielectric layer **112**, and one or more conformal layers **116** disposed on the gate dielectric layer **114**. The gate electrode layer **110** may be disposed on the conformal layers **116**. The interfacial dielectric layer **112** may include a dielectric material, such as an oxygen-containing material or a nitrogen-containing material, or multilayers thereof, and may be formed by any suitable deposition method, such as CVD, PECVD, or ALD. The gate dielectric layer **114** may include a dielectric material such as an oxygen-containing material or a nitrogen-containing material, a high-k dielectric material having a k value greater than that of silicon dioxide, or multilayers thereof. The gate dielectric layer **114** may be formed by any suitable method, such as CVD, PECVD, or ALD. The conformal layers **116** may include one or more barrier layers and/or capping layers, such as a nitrogen-containing material, for example tantalum nitride (TaN), titanium nitride (TiN), or the like. The conformal layers **116** may further include one or more work-function layers, such as aluminum titanium carbide, aluminum titanium oxide, aluminum titanium nitride, or the like. The term “conformal” may be used herein for ease of description upon a layer having substantial same thickness over various regions. The conformal layers **116** may be deposited by ALD, PECVD, MBD, or any suitable deposition technique.

[0020] One or more gate spacers **118** are formed along sidewalls of the gate stack **106** (e.g., sidewalls of the gate dielectric layers **114**). The gate spacers **118** may include silicon oxycarbide, silicon nitride, silicon oxynitride, silicon carbon nitride, the like, multi-layers thereof, or a combination thereof, and may be deposited by CVD, PVD, ALD, or other suitable deposition technique.

[0021] Portions of the gate stacks **106** and the gate spacers **118** may be formed on isolation regions **103**. The isolation regions **103** are formed on the substrate **101**. The isolation regions **103** may include an insulating material such as an oxygen-containing material, a nitrogen-containing material, or a combination thereof. The insulating material may be formed by a high-density plasma chemical vapor deposition (HDP-CVD), a flowable chemical vapor deposition (FCVD), or other suitable deposition process. In one aspect, the isolation regions **103** includes silicon oxide that is formed by a FCVD process.

[0022] A contact etch stop layer (CESL) **124** is formed on a portion of the S/D regions **104** and the isolation region **103**, and a first interlayer dielectric (ILD) **126** is formed on the CESL **124**. The CESL **124** can provide a mechanism to stop an etch process when forming openings in the first ILD **126**. The CESL **124** may be conformally deposited on surfaces of the S/D regions **104** and the isolation regions **103**. The CESL **124** may include an oxygen-containing material or a nitrogen-containing material, such as silicon nitride, silicon carbon nitride, silicon oxynitride, carbon nitride, silicon oxide, silicon carbon oxide, or the like, or a combination thereof, and may be deposited by CVD, PECVD, PVD, ALD, or any suitable deposition technique. The first ILD **126** may include an oxide formed by tetraethylorthosilicate (TEOS), undoped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), organosilicate glass (OSG), SiOC, and/or any suitable low-k dielectric materials (e.g., a material having a

dielectric constant lower than silicon dioxide), and may be deposited by spin-on, CVD, FCVD, PECVD, PVD, or any suitable deposition technique.

[0023] A silicide layer **120** is formed on at least a portion of each S/D region **104**, as shown in FIGS. 2A and 2B. The silicide layer **120** may include a material having one or more of WSi, CoSi, NiSi, TiSi, MoSi and TaSi. In some embodiments, the silicide layer **120** includes a metal or metal alloy silicide, and the metal includes a noble metal, a refractory metal, a rare earth metal, alloys thereof, or combinations thereof. A conductive contact **122** is disposed on each silicide layer **120**. The conductive contact **122** may include a material having one or more of Ru, Mo, Co, Ni, W, Ti, Ta, Cu, Al, TiN or TaN, and the conductive contact **122** may be formed by any suitable method, such as electro-chemical plating (ECP), or PVD. The silicide layer **120** and the conductive contact **122** may be formed by first forming an opening in the first ILD **126** and the CESL **124** to expose at least a portion of the S/D region **104**, then forming the silicide layer **120** on the exposed portion of the S/D region **104**, and then forming conductive contact **122** on the silicide layer **120**.

[0024] FIGS. 3A-3L are cross-sectional side views of various stages of manufacturing a semiconductor structure **300**, including an interconnection structure **301**, in accordance with some embodiments. In some embodiments, the interconnection structure **301** may be formed on or below the semiconductor device structure **100**. FIG. 5 is a flow chart of a method **500** for manufacturing the interconnection structure **301** in accordance with some embodiments. For the purpose of better describing the present disclosure, the cross-sectional side view of the semiconductor structure **300** in FIGS. 3A-3L and the method **500** in FIG. 5 will be discussed together. It is understood that the operations shown in the method **500** are not exhaustive and that other operations may be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIGS. 3A-3L and FIG. 5.

[0025] As shown in FIG. 3A, a semiconductor substrate **302** is provided. The semiconductor substrate **302** may be similar to substrate **101** discussed above. A plurality of devices **304** and a middle end of the line (MEOL) structure **306** may be formed on the semiconductor substrate **302**. In some embodiments, the plurality of devices **304** may be the devices **102** shown in FIGS. 2A and 2B.

[0026] In the MEOL structure **306**, low level interconnects (contacts), such as the conductive contacts **122** shown in FIGS. 2A and 2B, are formed over the S/D regions **104** and the gate electrode layer **110**. The MEOL structure **306** may have smaller critical dimensions and may be spaced closer together compared to a later formed back end of the line (BEOL) counterparts. A purpose of the contact layers of the MEOL structure **306** is to electrically connect the various regions of the transistors, i.e., the source/drain and metal gate electrode, to higher level interconnects in the BEOL.

[0027] As shown in FIG. 3A and operation **502** in FIG. 5, a conductive layer **308** is formed over the MEOL structure **306**, an etch stop layer (ESL) **310** may be formed over the conductive layer **308**, and a dielectric layer **312** is formed over the ESL **310**. In some embodiments, the conductive layer **308** may be a conductive layer of other interconnection structures of the semiconductor structure **300**. In some embodiments, the conductive layer **308** may be a conductive

layer above the MEOL structure **306**. In some embodiments, the conductive layer **308** may include Cu, Al, CuAl, Ru, Mo, W, and related alloys formed in a dielectric material (not shown). In some embodiments, the conductive layer **308** may be formed by ALD, CVD, PVD, electroless deposition (ELD), ECP, or other suitable processes.

[0028] In some embodiments, the ESL **310** may be used to control the etching depth in the dielectric layer **312** and serve as an etch stop when forming a later formed conductive feature in the dielectric layer **312**. In some embodiments, the ESL **310** may include SiNx, SiCxNy, AlNx, AlOx, AOxNy, SiOx, SiCx, SiOxCy, or other suitable materials. In some embodiments, the ESL **310** may be formed by CVD, PVD, ALD, spin coating, or other suitable processes. In some embodiments, the dielectric layer **312** may include or be made of porous SiCOH, dense SiCOH, BN, BC, or other suitable materials. In some embodiments, the dielectric layer **312** may be formed by PECVD, ALD, PVD, or other suitable processes.

[0029] As shown in FIG. 3B and operation **504** in FIG. 5, a first opening **314** and a second opening **316** are formed in the dielectric layer **312**. The first opening **314** is formed in the dielectric layer **312**, and the second opening **316** penetrates the dielectric layer **312** to expose a portion of the conductive layer **308**. In some embodiments, the first opening **314** and the second opening **316** may be formed by dry etch, wet etch, or other suitable processes. In some embodiments, the first opening **314** may include a trench formed in the dielectric layer **312**. In some embodiments, the second opening **316** may include a via and a trench formed sequentially in the dielectric layer **312** to expose a portion of the conductive layer **308**. In some embodiments, the trench of the first opening **314** and the trench of the second opening **316** may be formed first, and then the via of the second opening **316** may be formed under the trench of the second opening **316** to expose a portion of the conductive layer **308**. In some embodiments, the via of the second opening **316** may be formed first penetrating the dielectric layer **312** and the ESL **310** to expose a portion of the conductive layer **308**, and the trench of the first opening **314** and the trench of the second opening **316** may be formed thereafter.

[0030] As shown in FIG. 3C and operation **506** in FIG. 5, a liner layer **318** is conformally formed over the dielectric layer **312**. The liner layer **318** may cover the top surface of the dielectric layer **312**, the first opening **314**, and the second opening **316** including the exposed surface of the conductive layer **308**. In some embodiments, the liner layer **318** may include metal oxide, metal nitride, silicon oxide doped carbide (ODS), or other suitable materials. In some embodiments, the liner layer **318** may include AlOx, ZrOx, YOx, AlNx, TiNx, SiNx, SiCxNy, ODS, or other suitable materials. In some embodiments, the liner layer **318** may have a thickness between 5 Angstroms and 40 Angstroms. In some embodiments, the liner layer **318** may be formed by PECVD, ALD, PVD, or other suitable processes. The liner layer **318** may prevent the damage on sidewalls of the later formed barrier layer or conductive materials during a later etch process.

[0031] As shown in FIG. 3D and operation **508** in FIG. 5, a portion of the liner layer **318** that covers the conductive layer **308** is removed. In other words, the liner layer **318** at the bottom of the opening **316** is removed to expose the conductive layer **308**. In some embodiments, the removal of

the portion of the liner layer 318 may be performed by dry etch, or other suitable processes.

[0032] Then, as shown in FIG. 3E and operation 510 in FIG. 5, a barrier layer 320 is deposited over the liner layer 318 and the exposed conductive layer 308. In some embodiments, the barrier layer 320 may be conformally formed over the liner layer 318. In other words, the barrier layer 320 may cover not only the liner layer 318 but also the exposed conductive layer 308 at the bottom of the second opening 316. In some embodiments, the barrier layer 320 may include TaN, TiN, or other suitable materials. In some embodiments, the barrier layer 320 may have a thickness between 10 Angstroms and 30 Angstroms. In some embodiments, the barrier layer 320 may be formed by thermal ALD, or other suitable processes.

[0033] Then, as shown in FIG. 3E and operation 512 in FIG. 5, a first conductive feature 322 is formed in the first opening 314, and a second conductive feature 326 is formed in the second opening 316. A conductive material may be deposited over the barrier layer 320 and fills the first opening 314 and the second opening 316. Then, a planarization operation, e.g., chemical mechanical polishing (CMP), may be performed so that the first conductive feature 322 and the second conductive feature 326 are formed, as shown in FIG. 3E.

[0034] In some embodiments, the first conductive feature 322 and the second conductive feature 326 may include Cu, Al, CuAl, Ru, Mo, W, and related alloys. In some embodiments, the first conductive feature 322 and the second conductive feature 326 may be formed by ALD, CVD, PVD, ELD, ECP, or other suitable processes. The liner layer 318 is disposed between the first conductive feature 322 and the dielectric layer 312 and between the second conductive feature 326 and the dielectric layer 312.

[0035] Then, as shown in FIG. 3F, a capping layer 324 is formed on the first conductive feature 322 and the second conductive feature 326. In some embodiments, the capping layer 324 may include Cobalt (Co), or other suitable materials. In some embodiments, the capping layer 324 may be formed by CVD, ALD, or other suitable processes.

[0036] In some embodiments, the capping layer 324 is selectively formed on the first conductive feature 322 and the second conductive feature 326 but not on the dielectric layer 312. In some embodiments, before the formation of the capping layer 324, a pretreatment operation may be performed to clean the surfaces of the first conductive feature 322 and the second conductive feature 326. For example, a wet clean process may be performed to remove copper oxide on top surfaces of the first conductive feature 322 and the second conductive feature 326, some post CMP residue on the dielectric layer 312, and/or organic contamination from the CMP on the dielectric layer 312, the first conductive feature 322 and the second conductive feature 326.

[0037] In some embodiments, the capping layer 324 may be formed by CVD process with Co precursor and H₂. In some embodiments, the first conductive feature 322 and the second conductive feature 326 may include copper (Cu). For example, during the CVD process, H₂ strips the dicarbonyl groups from the Co precursor resulting in cobaltocene plus H₂. The Cu surfaces of the first conductive feature 322 and the second conductive feature 326 then bond with the hydrogen. Then, the cobaltocene replaces the hydrogen on the surfaces of the first conductive feature 322 and the second conductive feature 326 and forms Co capping layer

(the capping layer 324) on the first conductive feature 322 and the second conductive feature 326. In some embodiments, the capping layer 324 may be formed by CVD, ALD, or other suitable processes.

[0038] As shown in FIG. 3G and operation 514 in FIG. 5, a blocking layer 328 is formed on the dielectric layer 312. In some embodiments, the blocking layer 328 is formed by molecules with silicon-based function groups, and therefore the blocking layer 328 is formed on the dielectric layer 312, e.g., low k materials, but not on the capping layer 324, e.g., Co. For example, the blocking layer 328 may include a head group connected to a function group by way of a molecular chain. The head group is configured to adhere to preferred surfaces such as the surface of the dielectric layer 312 while not adhering to other surfaces such as the surfaces of the capping layer 324. In some embodiments, the head group may include butyltriethoxysilane, cyclohexyltrimethoxysilane, cyclopentyltrimethoxysilane, dodecyltriethoxysilane, dodecyltrimethoxysilane, decyltriethoxysilane, dimethoxy (methyl)-n-octylsilane, triethoxyethylsilane, ethyltrimethoxysilane, hexyltrimethoxysilane, hexyltriethoxysilane, hexadecyltrimethoxysilane, hexadecyltriethoxysilane, triethoxymethylsilane, trimethoxy(methyl) silane, methoxy (dimethyl) octadecylsilane, methoxy (dimethyl)-n-octylsilane, octadecyltriethoxysilane, triethoxy-n-octylsilane, octadecyltrimethoxysilane, trimethoxy (propyl) silane, trimethoxy-n-octylsilane, triethoxy (propyl) silane, methane, ethane, propane, butane, pentane, hexane, heptane, octane, nonane, decane, undecane, dodecane, pentadecane, hexadecane, any combination of the foregoing, or the like. In some embodiments, the function group may include a hydrophobic interfacial property that repels dielectric material, thereby preventing dielectric material from adhering to the blocking layer 328, in a later dielectric on metal (DOM) process. In some embodiments, the function group may include a methyl group, which provides the hydrophobic interfacial property. In some embodiments, the blocking layer 328 may be formed by a wet process, such as dip coating, spin coating, spraying coating, or other suitable processes.

[0039] As shown in FIG. 3H and operation 516 in FIG. 5, a dielectric layer 330 is formed on the capping layer 324. In some embodiments, the dielectric layer 330 may include metal oxide, metal nitride, or other suitable materials. In some embodiments, the dielectric layer 330 may be formed by thermal ALD, or other suitable processes. In some embodiments, the dielectric layer 330 may be formed by DoM selective deposition. As described above, because of the function group of the blocking layer 328 prevents dielectric material from adhering to the blocking layer 328. The dielectric layer 330 is formed only on the capping layer 324. In some embodiments, the dielectric layer 330 may prevent damage to the below layers, such as the capping layer 324, the first conductive feature 322, and the second conductive feature 326, during subsequent processing steps.

[0040] As shown in FIG. 3I, the blocking layer 328 is removed and an ESL 332 is conformally formed over the dielectric layer 312, the liner layer 318, the barrier layer 320, and the dielectric layer 330. In some embodiments, the ESL 332 may be formed by PVD, CVD, PECVD, ALD, plasma enhanced ALD (PEALD), or other suitable processes. In some embodiments, the ESL 332 may include silicon oxycarbide, silicon carbon nitride, silicon nitride, silicon carbon oxynitride, silicon dioxide, silicon carbide, silicon oxyni-

tride, aluminum nitride, aluminum oxynitride, aluminum oxide, another dielectric material, or other suitable materials. In some embodiments, the ESL 332 may have a thickness between 5 Angstroms and 200 Angstroms. Then, as shown in FIG. 3J, a portion of the ESL 332 is removed to expose portions of the dielectric layer 312.

[0041] As shown in FIG. 3K, a third opening 334 is formed in the dielectric layer 312 between the first conductive feature 322 and the second conductive feature 326. In some embodiments, an etch operation is performed on the ESL 332, the dielectric layer 312, the liner layer 318, the barrier layer 320, and the dielectric layer 330. In some embodiments, the etch operation may include dry etch, wet etch, or other suitable processes. Because the sidewalls of the barrier layer 320 are covered and protected by the liner layer 318, the liner layer 318 can prevent the damage on sidewalls of the barrier layer 320 during the etch operation.

[0042] As shown in FIG. 3L, a dielectric layer 336 is formed over the third opening 334 and an air gap 338 is formed between the first conductive feature 322 and the second conductive feature 326 in the dielectric layer 336. In some embodiments, the dielectric layer 336 may partially fill the third opening 334 resulting the air gaps 338. In some embodiments, the dielectric layer 336 may be the same material as the dielectric layer 312. In some embodiments, the dielectric layer 336 may include or be made of porous SiCOH, dense SiCOH, boron nitride (BN), boron carbide (BC), or other suitable materials. In some embodiments, the dielectric layer 336 may be formed by PECVD, ALD, PVD, or other suitable processes.

[0043] In some embodiments, for forming the air gap 338 in the dielectric layer 336, a non-conformal deposition process may be performed to form the dielectric layer 336 in the third opening 334 during operation 516. For example, the PECVD process may be performed to form the dielectric layer 336 in the third opening 334 and form the air gap 338 in the dielectric layer 336. In some embodiments, because of the deposition process is non-conformal, the air gap 338 may be triangle shaped or like triangle shaped, as shown in FIG. 3L. As shown in FIG. 3L, the air gap 338 is wider near a lower portion of the third opening 334 and narrower near an upper portion of the third opening 334. In some embodiments, the air gap 338 is defined by the dielectric layer 336 and the dielectric layer 312.

[0044] In some embodiments, the air gap 338 may reduce an effective dielectric constant of the dielectric layer 336. In some embodiments, the effective dielectric constant of the dielectric layer 336 may be reduced to a range between 2 and 3.6. By reducing the effective dielectric constant of the dielectric layer 336, the capacitance between the first conductive feature 322 and the second conductive feature 326 is reduced, and thereby the performance of the semiconductor structure 300 may be increased.

[0045] FIGS. 4A-4G are cross-sectional side views of various stages of manufacturing another semiconductor structure 400, including an interconnection structure 401, in accordance with some embodiments. In some embodiments, the interconnection structure 401 may be formed on or below the semiconductor device structure 100. FIG. 6 is a flow chart of a method 600 for manufacturing the interconnection structure 401 in accordance with some embodiments. For the purpose of better describing the present disclosure, the cross-sectional side view of the semiconductor structure 400 in FIGS. 4A-4G and the method 600 in

FIG. 6 will be discussed together. It is understood that the operations shown in the method 600 are not exhaustive and that other operations may be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIGS. 4A-4G and FIG. 6.

[0046] As shown in FIG. 4A, the structure and manufacturing process of the semiconductor structure 400 are similar to those of the semiconductor structure 300 in FIG. 3D. In other words, the operations 602, 604, 606, and 608 in the method 600 may be similar to the operations 502, 504, 506, and 508 in the method 500. However, as shown in FIG. 4A and operation 610 in FIG. 6, after removing a portion of the liner layer 318 that covers the conductive layer 308, a blocking layer 402 is formed at the bottom of the second opening 316 on the exposed conductive layer 308.

[0047] In some embodiments, the blocking layer 402 is formed by molecules with sulfur(S) or phosphorus (P) function groups. For example, the blocking layer 402 may include a head group connected to a function group by way of a molecular chain. In some embodiments, the head group has a high affinity to the metal surface (e.g., the exposed conductive layer 308), and thus adhere and/or anchor to the exposed conductive layer 308 rather than the liner layer 318.

[0048] As shown in FIG. 4B and operation 612 in FIG. 6, a barrier layer 404 is formed over the liner layer 318 but not on the blocking layer 402. In some embodiments, since the blocking layer 402 is formed by molecules with S or P function groups, the S or P function group may repel the deposition of the barrier layer 404 on the blocking layer 402. In some embodiments, the barrier layer 404 may include TaN, TiN, or other suitable materials. In some embodiments, the barrier layer 404 may have a thickness between 10 Angstroms and 30 Angstroms. In some embodiments, the barrier layer 404 may be formed by thermal ALD, or other suitable processes.

[0049] As shown in FIG. 4C and operation 614 in FIG. 6, the blocking layer 402 is removed. Then, as shown in operation 616 in FIG. 6, a first conductive feature 406, a second conductive feature 408, and a capping layer 410 are formed in the first opening 314 and the second opening 316 over the barrier layer 404. In some embodiments, the materials and the manufacturing processes of the first conductive feature 406, the second conductive feature 408, and the capping layer 410 may be similar to the materials and the manufacturing processes of the first conductive feature 322, the second conductive feature 326, and the capping layer 324. Because the blocking layer 402 prevents the formation of the barrier layer 404 at the bottom of the second opening 316, after forming the second conductive feature 408, the second conductive feature 408 can in direct contact with the conductive layer 308. Therefore, the resistance between the via structure (the second conductive feature 408) and the conductive layer 308 may be reduced.

[0050] As shown in FIG. 4D, a dielectric layer 412 is formed on the capping layer 410. An ESL 414 is formed over the dielectric layer 312, the liner layer 318, the barrier layer 404, and the dielectric layer 412. A dielectric layer 416 is formed and an air gap 418 is formed between the first conductive feature 406 and the second conductive feature 408 in the dielectric layer 416. In some embodiments, the materials and the manufacturing processes of the dielectric layer 412, the ESL 414, the dielectric layer 416, and the air gap 418 may be similar to the materials and the manufac-

turing processes of the dielectric layer 330, the ESL 332, the dielectric layer 336, and the air gap 338 shown in FIGS. 3F-3L.

[0051] The air gap 418 may reduce an effective dielectric constant of the dielectric layer 416. In some embodiments, the effective dielectric constant of the dielectric layer 416 may be reduced to a range between 2 and 3.6. By reducing the effective dielectric constant of the dielectric layer 416, the capacitance between the first conductive feature 406 and the second conductive feature 408 is reduced, and thereby the performance of the semiconductor structure 400 may be increased.

[0052] In addition, the liner layer 318 may prevent the damage at the sidewalls of the barrier layer 404, or the first conductive feature 406 and the second conductive feature 408, during the etch process of forming the opening between the first conductive feature 406 and the second conductive feature 408. The dielectric layer 412 may prevent top side damage of the first conductive feature 406 and the second conductive feature 408 during the etch process of forming the opening between the first conductive feature 406 and the second conductive feature 408.

[0053] Furthermore, in the semiconductor structure 400, because the blocking layer 402 prevents the formation of the barrier layer 404 at the bottom of the second opening 316, the second conductive feature 408 can in direct contact with the conductive layer 308. Therefore, the resistance of the via structure may be reduced. The resistance-capacitance (RC) delay of the semiconductor structure may be further reduced.

[0054] FIG. 4E illustrates an example of the semiconductor structure 400. In some embodiments, the semiconductor structure 400 may include the semiconductor substrate 302. The device 304, such as a transistor shown in FIG. 4E, may be formed on the semiconductor substrate 302. The MEOL 306 may include one or more than one conductive structure, such as one or more than one conductive layer and via, in contact with the terminals of the device 304. The interconnection structure 401 is formed on the MEOL 306.

[0055] FIG. 4F illustrates another example of the semiconductor structure 400. A conductive feature 450 may be formed above the interconnection structure 401 and in electric contact with the second conductive feature 408. FIG. 4F illustrates a further example of the semiconductor structure 400. Another interconnection structure 452 may be further formed on the interconnection structure 401. Furthermore, the interconnection structure 452 may include the air gap, or without the air gap as shown in FIG. 4G.

[0056] FIG. 4H is schematic cross-sectional side view of the semiconductor device structure 100 in accordance with some embodiments. The semiconductor device structure 100 may include the device layer 102 formed on and in the substrate 302 and an interconnection structure 460 formed over the device layer 102. The interconnection structure 460 includes various conductive features, such as a first plurality of conductive features 464 and second plurality of conductive features 466, and an intermetal dielectric (IMD) layer 462 to separate and isolate various conductive features 464, 466. In some embodiments, the first plurality of conductive features 464 are conductive lines and the second plurality of conductive features 466 are conductive vias. The interconnection structure 460 includes multiple levels of the conductive features 464, and the conductive features 464 are arranged in each level to provide electrical paths to various devices layer 102 disposed below. The conductive features

466 provide vertical electrical routing from the device layer 200 to the conductive features 464 and between conductive features 464. For example, the bottom-most conductive features 466 of the interconnection structure 460 may be electrically connected to the conductive contacts disposed over the S/D regions 104 (FIG. 2A) and the gate electrode layer 110 (FIG. 1B). The conductive features 464 and conductive features 466 may be made from one or more electrically conductive materials, such as metal, metal alloy, metal nitride, or silicide. For example, the conductive features 464 and the conductive features 466 are made from copper, aluminum, aluminum copper alloy, titanium, titanium nitride, tantalum, tantalum nitride, titanium silicon nitride, zirconium, gold, silver, cobalt, nickel, tungsten, tungsten nitride, tungsten silicon nitride, platinum, chromium, molybdenum, hafnium, iridium, other suitable conductive material, or a combination thereof.

[0057] The IMD layer 462 includes one or more dielectric materials to provide isolation functions to various conductive features 464, 466. The IMD layer 462 may include multiple dielectric layers embedding multiple levels of conductive features 464, 466. The IMD layer 462 is made from a dielectric material, such as SiOx, SiOxCyHz, SiOCN, SiON, or SiOxCy, where x, y and z are integers or non-integers. In some embodiments, the IMD layer 462 includes a low-k dielectric material having a k value less than that of silicon dioxide.

[0058] In some embodiments, the conductive features 464 disposed in a level of the interconnection structure 460 are partially overlapping with respect to the x-axis, which is substantially parallel to a major surface of the substrate 302, as shown in FIG. 4H. A level of the interconnection structure 460 may be a layer of the IMD layer 462. The layers are sometimes referred to as M1, M2, . . . M10, M11, et, with M1 being closest to the device layer 102. The air gap 418 described above may be included in any layers in the interconnection structure 460. Pitch of the conductive features 464, 466 increases from a lower portion 470 to an upper portion 472. In some embodiments, air gaps, such as the air gaps 418, may be located in one or more metal layers in the lower portion 470, for example M1, M2, but not presented in metal layers located in the upper portion 472, for example M10, M11. In some embodiments, the top metal layer does not include air gap. The partially overlapping conductive features 464 are described in detail below. Additional materials, such as glue layers, etch stop layers, and barrier layers, may be included in the interconnection structure 460 but are not shown in FIG. 4H for clarity.

[0059] An embodiment is an interconnection structure. The interconnection structure includes a first dielectric layer, a first conductive feature, a first liner layer, a second conductive feature, a second liner layer, and an air gap. The first conductive feature is disposed in the first dielectric layer. The first liner layer is disposed between the first conductive feature and the first dielectric layer. The second conductive feature penetrates the first dielectric layer. The second liner layer is disposed between the second conductive feature and the first dielectric layer. The air gap is disposed in the first dielectric layer between the first liner layer and the second liner layer. The first liner layer and the second liner layer include metal oxide, metal nitride, or silicon oxide doped carbide.

[0060] Another embodiment is a method for forming an interconnection structure. A first dielectric layer is deposited

over a conductive layer. A first opening is formed in the first dielectric layer and a second opening penetrates the first dielectric layer to expose a portion of the conductive layer. A liner layer is formed over the first dielectric layer. A portion of the liner layer is removed to expose the conductive layer under the second opening. A barrier layer is deposited over the liner layer and the exposed conductive layer. A first conductive feature is formed in the first opening and a second conductive feature is formed in the second opening. A blocking layer is deposited on the first dielectric layer. A second dielectric layer is formed on the first conductive feature not covered by the blocking layer and a third dielectric layer is formed on the second conductive feature not covered by the blocking layer.

[0061] A further embodiment is a method for forming an interconnection structure. A first dielectric layer is formed over a conductive layer. A first opening is formed in the first dielectric layer and a second opening is formed penetrating the first dielectric layer to expose a portion of the conductive layer. A liner layer is formed over the first dielectric layer. A portion of the liner layer is removed to expose the conductive layer under the second opening. A blocking layer is deposited on the exposed conductive layer at the bottom of the second opening. A barrier layer is formed over the liner layer not covered by the blocking layer. The blocking layer is removed to expose the conductive layer at the bottom of the second opening. A first conductive feature is formed in the first opening over the barrier layer and a second conductive feature is formed in the second opening in direct contact with the conductive layer.

[0062] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

1. An interconnection structure, comprising:
 - a first dielectric layer;
 - a first conductive feature disposed in the first dielectric layer;
 - a first liner layer disposed between the first conductive feature and the first dielectric layer;
 - a second conductive feature penetrating the first dielectric layer;
 - a second liner layer disposed between the second conductive feature and the first dielectric layer; and
 - an air gap disposed in the first dielectric layer between the first liner layer and the second liner layer, wherein the first liner layer and the second liner layer comprise metal oxide, metal nitride, or silicon oxide doped carbide.
2. The interconnection structure of claim 1, further comprising:
 - a second dielectric layer disposed on the first conductive feature; and
 - a third dielectric layer disposed on the second conductive feature.

3. The interconnection structure of claim 2, further comprising:
 - an etch stop layer disposed over at least a portion of the second dielectric layer and without covering the third dielectric layer.
4. The interconnection structure of claim 2, further comprising:
 - a cap layer disposed between the second dielectric layer and the first conductive feature and between the third dielectric layer and the second conductive feature.
5. The interconnection structure of claim 1, further comprising:
 - a first barrier layer disposed between the first conductive feature and the first liner layer; and
 - a second barrier layer disposed between the second conductive feature and the second liner layer, wherein the first barrier layer and the first liner layer comprise different materials, and the second barrier layer and the second liner layer comprise different materials.
6. The interconnection structure of claim 1, wherein the air gap is triangle shaped.
7. An interconnection structure, comprising:
 - a conductive layer;
 - a first dielectric layer disposed over the conductive layer;
 - a conductive feature disposed in the first dielectric layer;
 - a first liner layer disposed between the conductive feature and the first dielectric layer;
 - a via structure penetrating the first dielectric layer and in direct contact with the conductive layer;
 - a second liner layer disposed between the via structure and the first dielectric layer; and
 - an air gap disposed in the first dielectric layer between the first liner layer and the second liner layer.
8. The interconnection structure of claim 7, further comprising:
 - a first etch stop layer disposed between the conductive layer and the first dielectric layer.
9. The interconnection structure of claim 7, further comprising:
 - a second dielectric layer disposed on the conductive feature; and
 - a third dielectric layer disposed on the via structure.
10. The interconnection structure of claim 9, further comprising:
 - an etch stop layer disposed over at least a portion of the second dielectric layer.
11. The interconnection structure of claim 9, further comprising:
 - a cap layer disposed between the second dielectric layer and the conductive feature and between the third dielectric layer and the via structure.
12. The interconnection structure of claim 7, further comprising:
 - a first barrier layer disposed between the conductive feature and the first liner layer; and
 - a second barrier layer disposed between the via structure and the second liner layer.
13. The interconnection structure of claim 7, wherein the air gap is triangle shaped.
14. The interconnection structure of claim 7, wherein the first liner layer comprises metal oxide, metal nitride or

silicon oxide doped carbide, and the second liner layer comprises metal oxide, metal nitride or silicon oxide doped carbide.

15. An interconnect structure, comprising:
a first dielectric layer;
a first conductive feature disposed in the first dielectric layer;
a first liner layer disposed between the first conductive feature and the first dielectric layer;
a second conductive feature penetrating the first dielectric layer;
a second liner layer disposed between the second conductive feature and the first dielectric layer;
a second dielectric layer disposed over the first dielectric layer and the first and second conductive features, wherein an air gap is disposed between the first conductive feature and the second conductive feature, and the air gap is defined by the first dielectric layer and the second dielectric layer.

16. The interconnect structure of claim **15**, wherein the air gap is wider near a lower portion and wider near an upper portion.

17. The interconnect structure of claim **15**, further comprising:

a first cap layer disposed between the second dielectric layer and the first conductive feature; and

a second cap layer disposed between the second dielectric layer and the second conductive feature.

18. The interconnect structure of claim **17**, further comprising:

an etch stop layer disposed between the first dielectric and the second dielectric layer, wherein the etch stop layer is disposed on a portion of the first conductive feature.

19. The interconnect structure of claim **17**, further comprising:

a first barrier layer disposed between the conductive feature and the first liner layer; and

a second barrier layer disposed between the conductive feature and the second liner layer, wherein the first barrier layer and the first liner layer comprise different materials, and the second barrier layer and the second liner layer comprise different materials.

20. The interconnection structure of claim **19**, wherein the first liner layer comprises metal oxide, metal nitride or silicon oxide doped carbide.

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